

**PIL 13K TEU NEW BUILDS HUDONG
ZHONGHUA SHIPYARD**

H1933A / H1934A / H1935A / H1936A / H1937A

OBP SPARE LIST – REV 4

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EMERSON**

46. NITROGEN GENERATOR SYSTEM – AIR PRODUCTS

47. FGSS HYDRAULIC VALVE REMOTE CONTROL SYSTEM – HANLA IMS

48. LNG TANK INSULATION – TI

49. PAINT – JOTUN

50. CARGO FIXED FITTING & LASHING – SEC

1. HATCH COVER – CSSC HUAHAI

(OBP SPARE)

ITEM	Quantity
RESTING PAD (Propad) 30*80*150	50
RESTING PAD (Propad) 30*80*200	50

2. MOORING WINCH AND WINDLASS – WMMP

(OBP SPARE)

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Mooring Winches & Windlass - OBP Spare List

序号serial number	代号code	名 称name	数量quantity
电气部分 Electrical part			
1	CL2-523C	指示灯 indicator light	1
2	CL2-523R	指示灯 indicator light	1
3	CL2-523G	指示灯 indicator light	1
4	MP1-30G	按钮头 Button head	1
5	MP1-30R	按钮头 Button head	1
6	M2SS2-30B	选择开关头部 Selector switch head	1
机械部分 Mechanical part			
1	滚筒用刹车片 Roller brake pads		1套
2	齿轮箱无骨架油封 Gearbox frameless oil seal		2件
3	油脂嘴 Grease mouth		20件
液压部分 Hydraulic part			
1	o-ring	67296	1
2	o-ring	67827	7
3	o-ring	67950	4
4	o-ring	68082	1
5	o-ring	85347	1
6	o-ring	21290-2016B 301	1
7	o-ring	21290-2016B 304	1
8	o-ring	Y439400B 4	1
9	o-ring	Y439400B 7	1
10	o-ring	Y235562C 162	1
11	o-ring	Y235562C 265	1
12	o-ring	Y235562C 263	1
13	o-ring	Y235562C 264	1
14	o-ring	Y235562C 761	1
15	o-ring	Y235562C 762	1
16	o-ring	Y235562C 963	1
17	o-ring	Y40319799A 361	1
18	o-ring	Y40319799A 161	1
19	o-ring	Y40319799A 362	1
20	o-ring	Y40319799A 363	1
21	o-ring	107	1
22	o-ring	507	1
23	o-ring	508	1
24	o-ring	123	1
25	o-ring	115	1
26	o-ring	Y40325806 6	1
27	o-ring	Y40325806 7	1
28	o-ring	Y40325806 9	1
29	o-ring	Y40331013 126A	1
30	o-ring	Y40331013 126B	1
31	o-ring	Y40331013 125	1
32	o-ring	P702	1
33	o-ring	P703	1
34	o-ring	P710	1
35	o-ring	R702	1
36	o-ring	P719	1
37	o-ring	P728	1
38	o-ring	P729	1
39	o-ring	P706	1

40	o-ring	P774	1
41	o-ring	P792	1
42	o-ring	P703	1
43	o-ring	P704	1
44	o-ring	P705	1
45	o-ring	R707	1
46	o-ring	02380-9068 17-3	1
47	o-ring	02380-9068 17-5	1
48	o-ring	02380-9068 17-6	1
49	o-ring	02380-9068 17-8	1
50	o-ring	20	1
51	o-ring	21	2
52	o-ring	17	1
53	o-ring	Y40326798 126A	1
54	o-ring	Y40326798 126B	1
55	o-ring	Y40326798 125	1
56	o-ring	Y40322323	1
57	o-ring	Y40322323	1
58	o-ring	PTCZ102-1MV	1
59	o-ring	67825	1

3. E/R CRANE & PROVISION CRANE – KAIRONG

(OBP SPARE)

15T*7.965M E/R Crane Spare Parts List as OB

item	specification	QTY/shipset
轴承 Bearing	6211	2
电缆滑车 Pulley	HC041	2
销式油嘴 Pin type grease nipple	A-PT1/8	2
熔芯 Fuse Core	R054(500V,4A)	4
接触器 Contactor	LC1-D12B7C	1
接触器 Contactor	LC1-D09B7C	1
热继电器 Relay	LRD-08C (2.5-4A)	1
热继电器 Relay	LRD-10C (4-6A)	1

4T Provision Crane Spare Parts List as OB

item	specification	QTY/shipset
油嘴 Grease Nipple	A-PT1/8	5
断路器 Breaker	GV2-PM16C	1
熔芯 Fuse Core	R054(500V,4A)	2
指示灯 Lamp	AD16-22D/Y31S	1
指示灯 Lamp	AD16-22D/W23S	1

4. LIFEBOAT & RESCUE BOAT – JIANGSU BEIHAI

(OBP SPARE)

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Life Boat & Rescue Boat - OBP Spare List

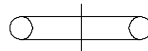
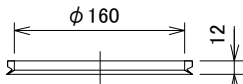
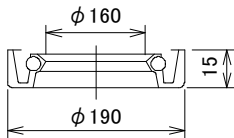
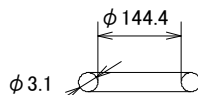
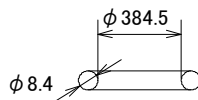
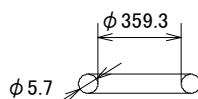
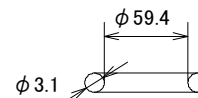
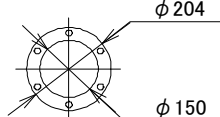
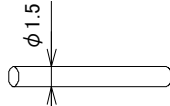
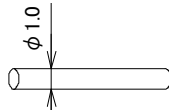
S/N	Equipment	Spare	Qty Per Ship	Remarks
1	LifeBoat	Hook Release Cable (Each type)	1 SET	
2		Engine Starter Motor	1 SET	
3		Canopy Light	1 PC	

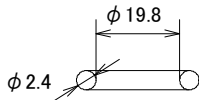
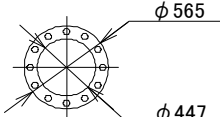
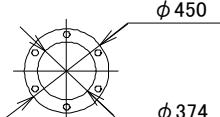
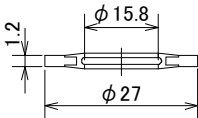
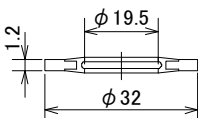
5. BOW THRUSTER – NAKASHIMA

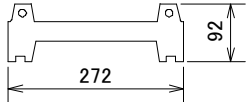
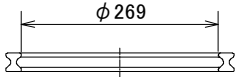
(OBP SPARE) – TO BE DELIVERED ON H1933A
ONLY

WORK NO.	9126/9127/9128/9129/9130	BIBLIOGRAPHY	
		OWNER	PACIFIC INTERNATIONAL LINES
		SHIPYARD	HUDONG-ZHONGHUA SHIPBUILDING (GROUP) CO., LTD.
		SNO.	H1933A/H1934A/H1935A/H1936A/H1937A
		CLASS	LR
		NT-C120 BOW THRUSTER SPECIAL SPARE PARTS LIST for SHIP OWNER	
(5 SHEETS WITH COVER)		 NAKASHIMA PROPELLER CO. , LTD. PROPULSION SYSTEM DESIGN DEPT.	
		APPROVED	<i>T. Mishima</i>
		CHECKED	<i>Y. Tanaka</i>
		DRAWN	<i>H. Yamaguchi</i>
		TRACED	<i>H. Yamaguchi</i>
		DATE	31st. Jul. , 2025
		DWGNO.	

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SPARE				PAGE		1 / 3		
NT-C120				SHIP NO.		H1933A/34A/ 35A/36A/37A		
SPECIAL SPARE PARTS LIST				BOX NO.				
NO.	NAME	SKETCH	MATE- RIAL	SUPPLY PER SHIP		DRAWING		REMARKS
				WORK -ING	SPA -RE	NO.	PART NO.	
1	BLADE SEAL ブレードシール		SPECIAL RUBBER	4	1	YP0000 -***	30	
2	V SEAL Vシール		NITRIL RUBBER	1	1	do	81	VR160A
3	OIL SEAL オイルシール		NITRIL RUBBER	2	2	do	82	AC4712E0
4	O - RING O - リング		NITRIL RUBBER	1	1	do	83	G145
5	O - RING O - リング		NITRIL RUBBER	1	1	do	85	P385
6	O - RING O - リング		NITRIL RUBBER	1	1	do	86	G360
7	O - RING O - リング		NITRIL RUBBER	24	6	do	91	G58
8	PACKING パッキン		SHEET PACKING	1	1	do	99	
9	LOCKING WIRE ロックングワイヤ		STAINLESS STEEL	1	1	do	135	30m
10	LOCKING WIRE ロックングワイヤ		STAINLESS STEEL	1	1	do	136	3m
MFR'S NAME & ADDRESS		NAKASHIMA PROPELLER CO. ,LTD. 688-1,JOTO-KITAGATA ,HIGASHI-KU ,OKAYAMA , 709-0625,JAPAN						

SPARE				PAGE		2 / 3		
NT-C120				SHIP NO.		H1933A/34A/ 35A/36A/37A		
SPECIAL SPARE PARTS LIST				BOX NO.				
NO.	NAME	SKETCH	MATE- RIAL	SUPPLY PER SHIP		DRAWING		REMARKS
				WORK- ING	SPA- RE	NO.	PART NO.	
11	O - RING O - リング		SPECIAL RUBBER	1	1	YP0000 -***	201	
12	PACKING パッキン		STAINLESS STEEL/ NITRIL RUBBER	1	1	do	11 -23	t=1
13	PACKING パッキン		SHEET PACKING	1	1	do	11 -25	t=1
14	SEAL WASHER シールワッシャ		STAINLESS STEEL/ NITRIL RUBBER	12	12	do	11 -28	M16用
15	SEAL WASHER シールワッシャ		STAINLESS STEEL/ NITRIL RUBBER	12	12	do	11 -29	M20用
MFR'S NAME & ADDRESS		NAKASHIMA PROPELLER CO. ,LTD. 688-1,JOTO-KITAGATA ,HIGASHI-KU ,OKAYAMA , 709-0625,JAPAN						

SPARE					PAGE		3 / 3	
NT-C120					SHIP NO.		H1933A/34A/ 35A/36A/37A	
SPECIAL SPARE PARTS LIST					BOX NO.			
NO.	NAME	SKETCH	MATE- RIAL	SUPPLY PER SHIP		DRAWING		REMARKS
				WORK -ING	SPA -RE	NO.	PART NO.	
1	GASKET ガスケット		FIBER FLEX	2	2	YP6000 -***	605	COUPLING 1150T10
2	SEAL RING シールリング		NITRIL RUBBER	2	2	do	606	COUPLING 1150T10
MFR'S NAME & ADDRESS		NAKASHIMA PROPELLER CO. ,LTD. 688-1,JOTO-KITAGATA ,HIGASHI-KU ,OKAYAMA , 709-0625,JAPAN						

6. ICCP & MGPS – KC LTD

(STANDARD + OBP SPARE)

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - ICCP & MGPS - Spare List

System	Name	Type	Quantity of Standard Spares	Quantity of OBP Spares
I.C.C.P System	AFT Anode Fuse	200A	1 PC	5 PCS
	FWD Anode Fuse	100A	1 PC	5 PCS
M.G.P.S	Main Fuse	4A	1 PC	5 PCS

7. FANS – HENYUAN MARINE EQUIPMENT

(OBP SPARE)

沪东船厂 PIL 13000TEU 1933A~1937A 风机OB备件清单

PIL 13K TEU Hudong Shipyard (H1933A-H1937A) - Cargo Hold Fans, Accommodation Fans & E/R Fans - OB Spare List

序号 Item	备件名称 Description	备件型号 Spare part model	数量 Qty.	单位 unit	备注 Remark
1	轴承 Bearings	6206-ZZ	6	个/船 pcs/ship	货仓 Cargo Hold Fans
2	轴承 Bearings	6306-ZZ	2	个/船 pcs/ship	货仓 Cargo Hold Fans
3	轴承 Bearings	6309-ZZ	22	个/船 pcs/ship	货仓 Cargo Hold Fans
4	轴承 Bearings	6311	112	个/船 pcs/ship	货仓 Cargo Hold Fans
5	轴承 Bearings	6204-ZZ	3	个/船 pcs/ship	机舱 E/R Fans
6	轴承 Bearings	6311-ZZ	1	个/船 pcs/ship	机舱 E/R Fans
7	轴承 Bearings	6314-ZZ	5	个/船 pcs/ship	机舱 E/R Fans
8	轴承 Bearings	6317-ZZ	5	个/船 pcs/ship	机舱 E/R Fans
9	叶轮 Impeller	JCZ-160T	1	个/船 pcs/ship	机舱 E/R Fans
10	轴承 Bearings	6204-ZZ	8	个/船 pcs/ship	房舱 Accommodation Fans
11	轴承 Bearings	6205-ZZ	2	个/船 pcs/ship	房舱 Accommodation Fans
12	轴承 Bearings	6206-ZZ	4	个/船 pcs/ship	房舱 Accommodation Fans
13	轴承 Bearings	6309-ZZ	8	个/船 pcs/ship	房舱 Accommodation Fans

上海亨远船舶设备有限公司

Shanghai Hengyuan Marine Equipment Co.,Ltd.

8. ELEVATOR – GSIE

(OBP SPARE)

MCE 船用电梯 Marine Elevator

备品备件清单

APARE PARTS LIST

序号No.	名 称 Name	型号Type	数量Qty
1	按钮 BUTTON	MCTC00-AM-SC-05	10
2	显示板 DISPLAY BOARD	MCTC-HCB-D634S-GC	2
3	轿厢导靴靴衬 LININGS FOR GUIDE SHOUE(CAR)	DX20	6
4	对重导靴靴衬 LININGS FOR GUIDE SHOUE(CWT)	DX20	6
5	门滑块 DOOR SHOE	2L1A050062	10
6	平层开关 LEVELING SWITCH	SGD31-GG-TZ2B2G	2
7	门触点 DOOR CONTACT	T12-A1ZKS	5
8	运行接触器 KDY2 CONTACTOR	LC1D-25F7C AC110	1
	抱闸接触器 LZK CONTACTOR	LC1D-09F7C AC110	1
	封星接触器 KFX CONTACTOR	LC1D-258F7C AC110V	1
9	继电器 RELAY	RXM2LB2BD DC24V	1
10	行程开关 TRAVEL SWITCH	236-ZR11	2
11	急停开关 EMERGENCY STOP SWITCH	XB2-BS542C	1
12	选择开关 OPTION SWITCH	BLA-KH21-712-000	1
13	限位开关 LIMIT SWITCH	T4VH 336-11z-M20-1058	2
14	层门钥匙 LANDING DOOR KEY	三角钥匙	4
15	工具箱 TOOL CABINET		1

9. GALLEY & LAUNDRY EQUIPMENT – AKER GL

(OBP SPARE)



AKER MARINE & OFFSHORE (UK) LIMITED

Project No.:	Vessel Type: 13K TEU
Equipment: Galley & Laundry Equipment	Hull No.: H1933A-H1937A
Consignee: Hudong Zhonghua Shipyard	Offer No.:
Address:	

Spare Parts List

Item	Equipment	Spare Parts	Specification	Quantity
1	电灶+烤箱备件 SPARES FOR Range+oven ACRZ-30	断路器 Breaker	DZ47s C 3P-63A	2
		绿指示灯 Indicator Light White	AD16-22	2
		指示灯白 Indicator Light White	AD16-22	2
		断路器 Breaker	DZ47s C 3P-10A	2
		烤箱加热管 Oven Heating Tube	CRZ/18-02-4-1-0	2
2	多用机备件 SPARES FOR Marine Mixer AB30-E	启停按钮 Start-stop switch	AD11-22/41-9GZ-440V	1
3	可倾式电汤锅备件 SPARES FOR Tilting electric soup pot	断路器 Breaker	DZ47s C 3P-25A	1
		指示灯白 Indicator Light White	AD16-22	1
4	电烤箱 SPARES FOR Electric baking oven ARKX-5	指示灯白 Indicator Light White	AD11-22/41-9GZ-440V	1
		指示灯绿 Indicator Light Green	AD11-22/41-9GZ-440V	1
		加热管 Heating Tube	RKCB220/750	1
5	饮水机备件 SPARES FOR Water Dispenser ADL-2C	滤芯 Filter Element	10"UDF+10PP	2
6	保温台（干式）备件 SPARES FOR Warm Stage ARWT-60	指示灯白 Indicator Light White	AD16-22	1
		断路器 Breaker	DZ47s C 3P-10A	1
5	保温台（干式）备件 SPARES FOR Warm Stage ARWT-30	指示灯白 Indicator Light White	AD16-22	1
		断路器 Breaker	DZ47s C 3P-10A	1

10. DF MAIN ENGINE – CMD

(OBP SPARE)

11. MAIN ENGINE TURBOCHARGER – MHI

(OBP SPARE)

NOTE: ONLY TURBOCHARGER ROTOR
COMPLETE IS DELIVERED TO PIL
WAREHOUSE

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Main Engine Turbocharger - Spare List

S/N	DESCRIPTION	QTY / SHIPSET	REMARKS
1	Turbocharger rotor complete	1 (per series of vessel)	
2	Bearings (journal bearings and thrust bearings)	1	
3	ITEM 362: O-RING	1	Overhaul Kit C1
4	ITEM 373: O-RING	1	Overhaul Kit C1
5	ITEM 449: O-RING	1	Overhaul Kit C1
6	ITEM 472: O-RING	1	Overhaul Kit C1
7	ITEM 225: BOLT	1	Overhaul Kit C1
8	ITEM 226: DOUBLE TOUNGED WASHER	1	Overhaul Kit C1
9	ITEM 351: SPRING WASHER	1	Overhaul Kit C1
10	ITEM 354: O-RING	1	Overhaul Kit C1
11	ITEM 831: PACKING	1	Overhaul Kit C1
12	ITEM 871: PACKING	1	Overhaul Kit C1

12. DF AUXILLIARY ENGINE GENERATOR – WARTSILA CWEC

(OBP SPARE)

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Auxilliary Engine - OBP Spares		
S/N	DESCRIPTION	QTY / SHIPSET
1	Cylinder head complete include inlet/outlet valve	9
2	Cylinder liner	9
3	Piston	9
4	Piston pin complete	9
5	Retaining pin for piston	18
6	Piston ring set	9
7	Connecting rod, lower part	9
8	Connecting rod, upper part	9
9	Big end bearing Kit	9
10	Shim plate for Con-rod	9
11	Main bearing kit	9
12	Gas admission valve	9
13	Fuel injection pump	9
14	Plunger assembly for fuel injection pump	9
15	Fuel injection valve	9
16	Nozzle	9

13. DF AUXILLIARY ENGINE GENERATOR
TURBOCHARGER – ABB

(OBP SPARE) – DELIVERED TO PIL
WAREHOUSE

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Auxilliary Engine Turbocharger - OBP Spares		
S/N	DESCRIPTION	QTY / SHIPSET
1	Complete AE TC A150-M	1 PER SERIES

14. EMERGENCY GENERATOR - CMXD

(OBP SPARE)

Item	Spare Part Name	Qty/per ship
1	Element ,air cleaner	2
2	Fuel filter	2
3	Element, fuel filter	1
4	Lub.oil filter	1

4. We could provide EDG operation and maintenance training of your staff and crew in the shipyards if you have requirements.

Best Regards,

Lily Han 韩莉

Zhenjiang China Marine-Xiandai Generating Co.,Ltd.

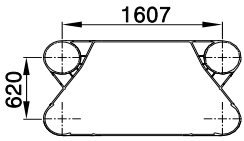
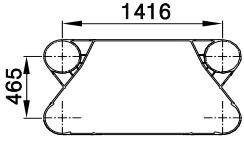
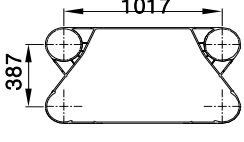
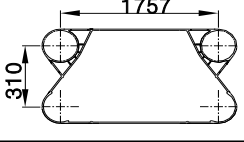
Title:Sales Manager of Sales Department

Mobile: +86 19602184638 E-mail: h15862983670@163.com

Address: Panzong Road, Hi-tech Garden, Zhenjiang City, Jiangsu Province 212009, China

15. PLATE COOLER – LHE

(OBP SPARE)

SPARE PART LIST				HULL NO.		
PLATE HEAT EXCHANGER				H1933A~H1934A		
NO.	NAME	SKETCH	MATERIAL	SUPPLY/SHIP		REMARK
				WORKING	SPARE	
1	GASKET (HT316)		NBR		92	
2	GASKET (SH232)		NBR		84	
3	GASKET (XT201D)		NBR		10	
4	GASKET (SH103)		NBR		8	
MANUFACTURER'S NAME & ADDRESS		 #1089 JANGBANG-RI, HALLIM-MYUN, GIMHAE-SI, GYUNGNAM, KOREA TEL : +82-55-340-0626, FAX : +82-55-340-0629				

16. AIR COMPRESSORS – JP SAUER

(OBP SPARE)

17. PUMPS – NANIWA

(STANDARD + OBP SPARE)

STANDARD SPARE PARTS + ADDITIONAL SPARE PARTS FOR SHIPOWNER
Hudong Zhonghua Shipbuilding, 13,000TEU Container Vessel, Hull No.H1933A/34A/35A/36A/37A

<u>(Parts name)</u>	<u>(Parts no.)</u>	<u>(Q'ty/ship)</u>	
For MAIN COOLING S.W. PUMP		Q'ty: 3 set(s)/ship	Model... FEWV-450
1) Bearing Metal	204	1 pce	
2) Mechanical Seal	520	1 set	
3) Mechanical Seal (for shipyard commissioning)	520	3 sets	
4) Casing Ring	020	1 pce	
5) Casing Ring	021	1 pce	
For BALLAST PUMP		Q'ty: 2 set(s)/ship	Model... FEWV-350-3
1) Bearing Metal	204	1 pce	
2) Mechanical Seal	520	1 set	
3) Mechanical Seal (for shipyard commissioning)	520	2 sets	
4) Casing Ring	020	1 pce	
5) Casing Ring	021	1 pce	
For BILGE, FIRE & G/S PUMP		Q'ty: 2 set(s)/ship	Model... FE2V-250-3
1) Ball Bearing	201	1 pce	
2) Mechanical Seal	520	1 set	
3) Submerged Bearing	203	1 pce	
4) Mechanical Seal (for shipyard commissioning)	520	2 sets	
5) Casing Ring	020	1 pce	
6) Casing Ring	021	1 pce	
7) Mechanical Seal	520	1 set	
8) Pressure Gauge		1 pce	
9) Compound Gauge		1 pce	
For L.T. COOLING F.W. PUMP		Q'ty: 2 set(s)/ship	Model... FEWV-400-3
1) Bearing Metal	204	1 pce	
2) Mechanical Seal	520	1 set	
For M/E AIR COOLER L.T. COOLING F.W. PUMP		Q'ty: 2 set(s)/ship	Model... FEWV-350
1) Bearing Metal	204	1 pce	
2) Mechanical Seal	520	1 set	
For M/E H.T.F.W. PUMP		Q'ty: 2 set(s)/ship	Model... FEV-200E1
For L/T. C.W. BOOSTER PUMP		Q'ty: 2 set(s)/ship	Model...
1) Bearing Metal	204	1 pce	
2) Mechanical Seal	520	1 set	
For M/E JACKET WATER PREHEATER PUMP		Q'ty: 1 set(s)/ship	Model... BHR-65
1) Mechanical Seal	520	1 set	
For EMERGENCY FIRE PUMP		Q'ty: 1 set(s)/ship	Model... FGV-100-2
1) Bearing Metal	204	1 pce	
2) Mechanical Seal	520	1 set	
3) Mechanical Seal (for shipyard commissioning)	520	1 set	
4) Casing Ring	020	1 pce	
5) Casing Ring	021	1 pce	
6) Mechanical Seal	520	1 set	
7) Pressure Gauge		1 pce	
8) Compound Gauge		1 pce	
For MAIN L.O. PUMP		Q'ty: 2 set(s)/ship	Model... TOM-300-2
1) Ball Bearing	201	1 pce	
2) Submerged Bearing	203	1 pce	
3) Oil Seal	530	1 set	
For FRESH WATER PUMP		Q'ty: 2 set(s)/ship	Model... BHR-50-3
1) Mechanical Seal	520	1 set	
For HOT WATER CIRCULATING PUMP		Q'ty: 2 set(s)/ship	Model... BHR-32
1) Mechanical Seal	520	1 set	
For CHEMICAL CLEANING PUMP		Q'ty: 1 set(s)/ship	Model... BHR-40
For NaOH TRANSFER PUMP		Q'ty: 2 set(s)/ship	Model...
1) Mechanical Seal	520	1 set	

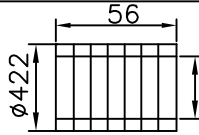
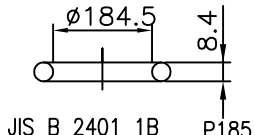
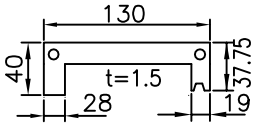
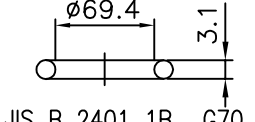
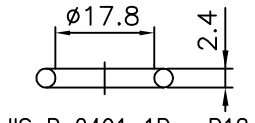
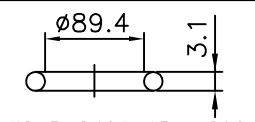
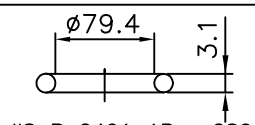
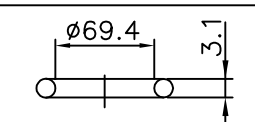
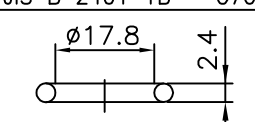
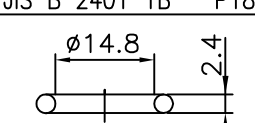
STANDARD SPARE PARTS + ADDITIONAL SPARE PARTS FOR SHIPOWNER
Hudong Zhonghua Shipbuilding, 13,000TEU Container Vessel, Hull No.H1933A/34A/35A/36A/37A

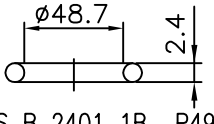
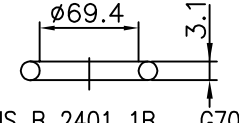
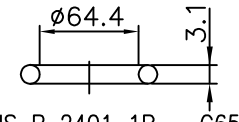
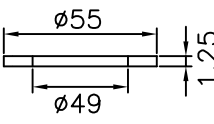
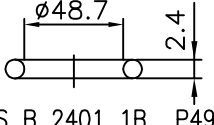
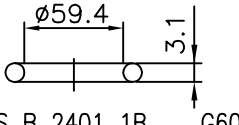
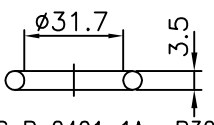
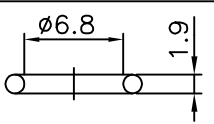
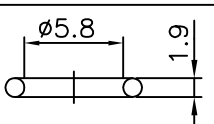
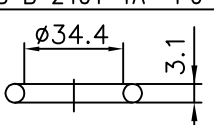
<u>(Parts name)</u>	<u>(Parts no.)</u>	<u>(Q'ty/ship)</u>	
For GREY WATER DISCHARGE PUMP		Q'ty: 1 set(s)/ship	Model... ANH-50
For SEWAGE DISCHARGE PUMP		Q'ty: 1 set(s)/ship	Model...
1) Ball Bearing	0020	1 pce	
2) Gland Packing	7040	1 set	
For E/R SEWAGE DISCHARGE PUMP		Q'ty: 1 set(s)/ship	Model... ANH-40
1) Ball Bearing	0020	1 pce	
2) Gland Packing	7040	1 set	
For HOT GLYCOL WATER CIRCULATING PUMP		Q'ty: 2 set(s)/ship	Model... BHR-100
1) Mechanical Seal	520	1 set	
For COLD GLYCOL WATER CIRCULATING PUMP		Q'ty: 2 set(s)/ship	Model... BHR-50
1) Mechanical Seal	520	1 set	
For C/H WATER SPRAY PUMP		Q'ty: 1 set(s)/ship	Model... FEWV-250-4D
1) Bearing Metal	204	1 pce	
2) Mechanical Seal	520	1 set	
4) Casing Ring	020	1 pce	
5) Casing Ring	021	1 pce	
For iCER COOLING SEA WATER PUMP		Q'ty: 2 set(s)/ship	Model... FEWV-350-3
1) Bearing Metal	204	1 pce	
2) Mechanical Seal	520	1 set	
4) Casing Ring	020	1 pce	
5) Casing Ring	021	1 pce	
For ULSHFO TRANSFER PUMP		Q'ty: 1 set(s)/ship	Model...
For FUEL OIL CONSTANT TRANSFER PUMP		Q'ty: 1 set(s)/ship	Model... ALTV-125
For MGO TRANSFER PUMP		Q'ty: 1 set(s)/ship	Model...
1) Ball Bearing	122	1 pce	
2) Mechanical Seal	509	1 set	
For M/E CROSSHEAD L.O. PUMP		Q'ty: 2 set(s)/ship	Model... AHSV-118N
1) Ball Bearing	103	1 pce	
2) Ball Bearing	105	1 pce	
3) Mechanical Seal	702	1 set	
For E/R BILGE TRANSFER PUMP		Q'ty: 1 set(s)/ship	Model... 2VP-10
For E/R CLEAN BILGE PUMP		Q'ty: 1 set(s)/ship	Model...
1) Gland Packing	501	1 set	
2) Ball Bearing	630	1 pce	
3) V-Belt	676	1 pce	
For SLUDGE PUMP		Q'ty: 1 set(s)/ship	Model... AE1E 750-ID
1) Ball Bearing	103	1 pce	
2) Ball Bearing	104	1 pce	
3) Gland Packing	207	1 set	
For LUBE OIL TRANSFER PUMP		Q'ty: 1 set(s)/ship	Model... ALG-65
1) Bearing Metal	201	1 pce	
2) Mechanical Seal	520	1 set	
For M/E CYLINDER OIL TRANSFER PUMP		Q'ty: 2 set(s)/ship	Model... ALG-32N
1) Bearing Metal	201	1 pce	
2) Mechanical Seal	520	1 set	
For M/E T/C L.O. PUMP		Q'ty: 2 set(s)/ship	Model... ALTV-90
1) Ball Bearing	122	1 pce	
2) Mechanical Seal	509	1 set	

18. STEERING GEAR – YOOWOON

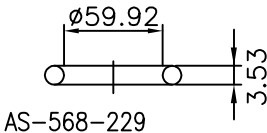
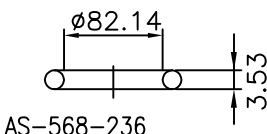
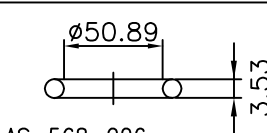
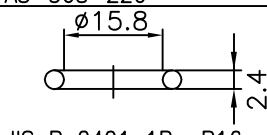
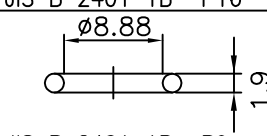
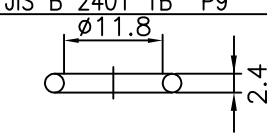
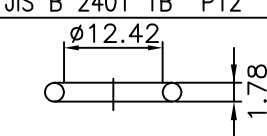
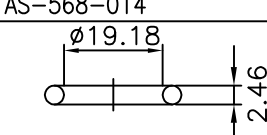
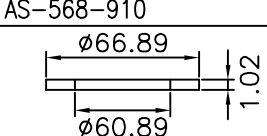
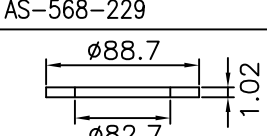
(STANDARD + OBP SPARE)

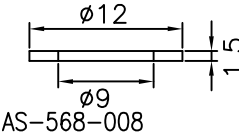
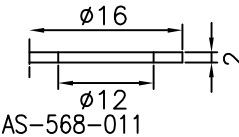
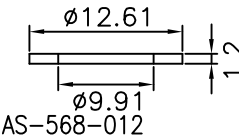
STANDARD + ADDITIONAL		12 sheets with cover	
		YDFT-760-3 ELEC. HYDR. STEERING GEAR LIST OF SPARE AND TOOL	
	APPROVED H.D.H		
	CHECKED K.D.H		
	DRAWN L.J.H		
CLASS	SCALE		
	SERIAL No.:	DRAWING No. YS-RDT1933-MS000-0	REV. NO.
	YOOWON INDUSTRIES LTD.		

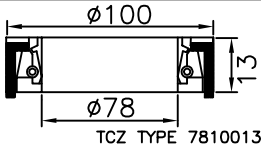
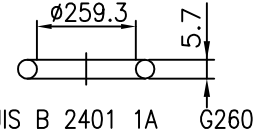
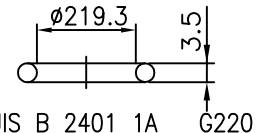
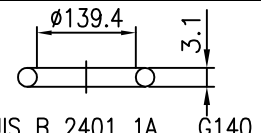
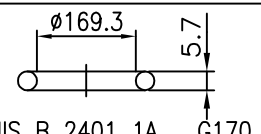
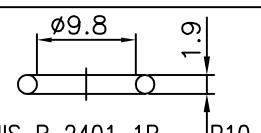
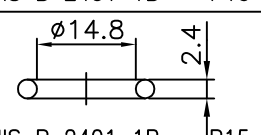
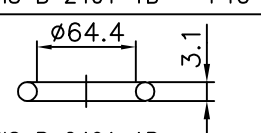
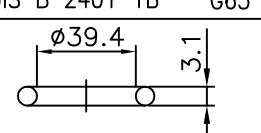
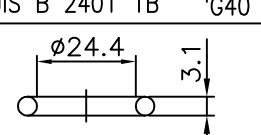
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YDFT-760							BOX No.			
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				WORK- ING	STANDARD SPARE	ADDITIONAL SPARE	No.	PART No.		
S-1	PACKING FOR OIL CYLINDER		SPECIAL PACKING	4 SETS	4 SETS	-	YS-CD08706 -00210-0	2-10	F-390 NV-A	
S-2	O RING FOR OIL CYLINDER		SYNTHETIC RUBBER	4	2	2	YS-CD08706 -00200-0	1-11	360-084 -185-2	
S-3	PACKING FOR CHAIN COUPLING		SYNTHETIC RUBBER	3	3	-	YS-CD08706 -00K00-0	K-1	NR-8022 -PK	
S-4	O RING FOR VALVE BLOCK		SYNTHETIC RUBBER	6	3	3	N76-V10 -0303	1	361-031 -070-2	
S-5	O RING FOR VALVE BLOCK		SYNTHETIC RUBBER	6	3	3	N76-V10 -0303 -0411	3	360-024 -018-2	
S-6	O RING FOR TRANSFER VALVE		SYNTHETIC RUBBER	3	2	2	N76-V10 -0299	21	361-031 -080-2	
S-7	O RING FOR MAN.ISO. VALVE		SYNTHETIC RUBBER	2	1	1	YS-VBTM602 -CSA00-0	12	361-031 -080-2	
S-8	O RING FOR TRANSFER VALVE AND MAN.ISO. VALVE		SYNTHETIC RUBBER	4	2	2	YS-VBTM602 -CSA00-0	5	361-031 -070-2	
S-9	O RING FOR TRANSFER VALVE		SYNTHETIC RUBBER	10	5	5	N76-V10 -0299	13	360-024 -018-2	
S-10	O RING FOR TRANSFER VALVE		SYNTHETIC RUBBER	3	2	2	N76-V10 -0299	14	361-024 -015-2	
MFR'S NAME & ADDRESS		YOOWON INDUSTRIES LTD.								

SPARE							PAGE		
ELECTRO-HYDRAULIC STEERING GEAR YDFT-760							SHIP No.		
							BOX No.		
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				WORK- ING	STANDARD SPARE	ADDITIONAL SPARE	No.	PART No.	
S-11	O RING FOR MANUAL ISO. VALVE	 JIS B 2401 1B P49	SYNTHETIC RUBBER	4	2	2	YS-VBTM602 -CSA00-0	11	360-024 -049-2
S-12	O RING FOR MANUAL ISO. VALVE	 JIS B 2401 1B G70	SYNTHETIC RUBBER	4	2	2	YS-VBTM602 -CSA00-0	13	361-031 -070-2
S-13	O RING FOR MANUAL ISO. VALVE	 JIS B 2401 1B G65	SYNTHETIC RUBBER	4	2	2	YS-VBTM602 -CSA00-0	15	361-031 -065-2
S-14	BACK UP RING FOR STOP VALVE & BY-PASS VALVE		SPECIAL RESIN	6	3	3	N76-V10 -0301	3	N76-367 -0002-3
S-15	O RING FOR STOP VALVE & BY-PASS VALVE	 JIS B 2401 1B P49	SYNTHETIC RUBBER	6	3	3	N76-V10 -0301	6	360-035 -049-2
S-16	O RING FOR STOP VALVE & BY-PASS VALVE	 JIS B 2401 1B G60	SYNTHETIC RUBBER	4	2	2	N76-V10 -0301	8	361-031 -030-2
S-17	O RING FOR AUTO. BY PASS VALVE	 JIS B 2401 1A P32	SYNTHETIC RUBBER	3	2	2	N76-V10 -0083	21	361-035 -032-2
S-18	O RING FOR AUTO. BY PASS VALVE	 JIS B 2401 1A P7	SYNTHETIC RUBBER	3	2	2	N76-V10 -0083	22	361-035 -007-2
S-19	O RING FOR AUTO. BY PASS VALVE	 JIS B 2401 1A P6	SYNTHETIC RUBBER	3	2	2	N76-V10 -0083	23	361-019 -006-1
S-20	O RING FOR AUTO. BY PASS VALVE	 JIS B 2401 1B G35	SYNTHETIC RUBBER	6	3	3	N76-V10 -0083	24	361-031 -035-2
MFR'S NAME & ADDRESS		YOOWON INDUSTRIES LTD.							

SPARE							PAGE		
ELECTRO-HYDRAULIC STEERING GEAR YDFT-760							SHIP No.		
							BOX No.		
NO	NAME OF PART	SKETCH	MATERIAL	SUPPLY PER SHIP			DRAWING		REMARKS
				WORK-ING	STANDARD SPARE	ADDITIONAL SPARE	No.	PART No.	
S-21	O RING FOR MANIFOLD BLOCK		SYNTHETIC RUBBER	3	2	2	N76-V10-0302	10	360-035-035-2
S-22	O RING FOR FLANGE (HIGH PRESS PIPING)		SYNTHETIC RUBBER	12	6	6		03	361-031-065-2
S-23	O RING FOR FLANGE (HIGH PRESS PIPING)		SYNTHETIC RUBBER	4	2	2		01	360-035-065-2
S-24	O RING FOR FLANGE (HIGH PRESS PIPING)		SYNTHETIC RUBBER	18	9	9		02	361-031-070-2
MFR'S NAME & ADDRESS		YOOWON INDUSTRIES LTD.							

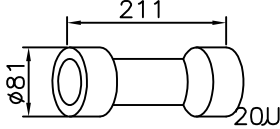
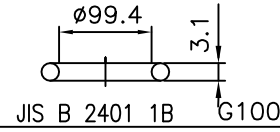
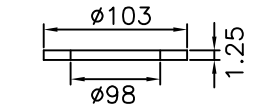
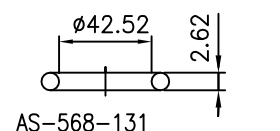
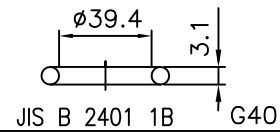
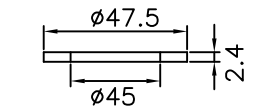
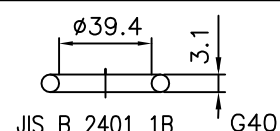
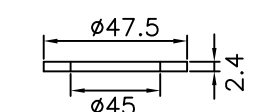
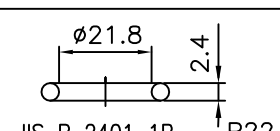
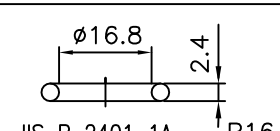
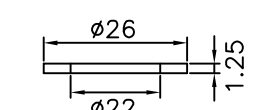
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ELECTRO-HYDRAULIC STEERING GEAR							SHIP No.		
LOGIC RELIEF VALVE(CE050)							BOX No.		
NO	NAME OF PART	SKETCH	MATERIAL	SUPPLY PER SHIP			DRAWING		REMARKS
				WORK- ING	STANDARD SPARE	ADDITIONAL SPARE	No.	PART No.	
V-1	O RING FOR LOGIC VALVE		NBR	8	2	2	2-229	8	
V-2	O RING FOR LOGIC VALVE		NBR	4	1	1	2-236	10	
V-3	O RING FOR RATAINER		NBR	4	1	1	2-226	12-1	
V-4	O RING FOR RELIEF VALVE		NBR	4	1	1	P-16	16B	
V-5	O RING FOR RELIEF VALVE		NBR	4	1	1	P-9	14B	
V-6	O RING FOR RELIEF VALVE		NBR	4	1	1	P-12	17B	
V-7	O RING FOR CHECK VALVE		NBR	4	1	1	AS568-014	20C	
V-8	O RING FOR CHECK VALVE		NBR	4	1	1	AS568-910	19C	
V-9	BACK UP RING FOR LOGIC VALVE		NBR	16	4	4	2-229	9	
V-10	BACK UP RING FOR LOGIC VALVE		NBR	4	1	1	8-236	11	
MFR'S NAME & ADDRESS		YOOWON INDUSTRIES LTD.							

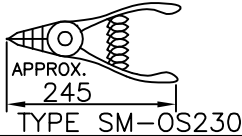
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ELECTRO-HYDRAULIC STEERING GEAR							SHIP No.		
LOGIC RELIEF VALVE							BOX No.		
NO	NAME OF PART	SKETCH	MATERIAL	SUPPLY PER SHIP			DRAWING		REMARKS
				WORK- ING	STANDARD SPARE	ADDITIONAL SPARE	No.	PART No.	
V-11	BACK UP RING FOR RELIEF VALVE	 AS-568-008	TEFLON	4	1	1	17	8-008	
V-12	BACK UP RING FOR RELIEF VALVE	 AS-568-011	TEFLON	4	1	1	19	8-011	
V-13	BACK UP RING FOR CHECK VALVE	 AS-568-012	TEFLON	4	1	1	22	8-012	
MFR'S NAME & ADDRESS		YOOWON INDUSTRIES LTD.							

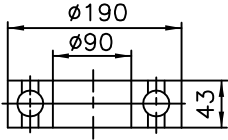
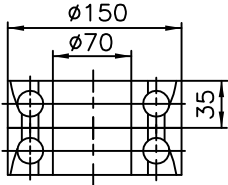
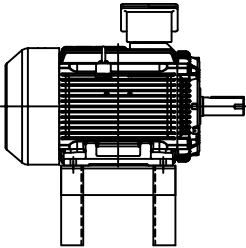
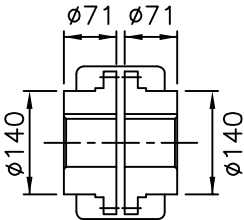
SPARE							PAGE		
6V-FH2B-MK-JANNEY PUMP							SHIP No.		
							BOX No.		
NO	NAME OF PART	SKETCH	MATERIAL	SUPPLY PER SHIP			DRAWING		REMARKS
				WORK- ING	STANDARD SPARE	ADDITIONAL SPARE	No.	PART No.	
PS-1	OIL SEAL		SYNTHETIC RUBBER	3	1	—	RA5500-211-0	36	
PS-2	O RING FOR FRONT PLATE		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	47	361-057-260-1
PS-3	O RING FOR END PLATE		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	48	361-057-220-1
PS-4	O RING FOR SEAL BOX		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	49	361-031-140-1
PS-5	O RING FOR CASING		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	50	361-057-170-1
PS-6	O RING		SYNTHETIC RUBBER	12	3	3	RA5500-211-0	51	360-019-010-2
PS-7	O RING FOR CASING		SYNTHETIC RUBBER	6	2	2	RA5500-211-0	52	360-024-015-2
PS-8	O RING FOR END PLATE		SYNTHETIC RUBBER	6	2	2	RA5500-211-0	53	360-031-065-2
PS-9	O RING FOR END PLATE		SYNTHETIC RUBBER	6	2	2	RA5500-211-0	54	360-031-040-2
PS-10	O RING		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	55	361-031-025-2
MFR'S NAME & ADDRESS		YOOWON INDUSTRIES LTD.							

SPARE							PAGE		
6V-FH2B-MK-JANNEY PUMP							SHIP No.		
							BOX No.		
NO	NAME OF PART	SKETCH	MATERIAL	SUPPLY PER SHIP			DRAWING		REMARKS
				WORK- ING	STANDARD SPARE	ADDITIONAL SPARE	No.	PART No.	
PS-11	O RING		SYNTHETIC RUBBER	6	2	2	RA5500-211-0	56	GA1150-831-0
PS-12	O RING FOR CASING		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	57	361-057-195-1
PS-13	O RING FOR CASING		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	58	361-031-135-1
PS-14	BACK UP RING		SPECIAL RESIN	12	4	4	RA5500-211-0	59	YA9103-167-0
PS-15	BACK UP RING		SPECIAL RESIN	6	2	2	RA5500-211-0	60	YA9103-017-0
PS-16	BACK UP RING		SPECIAL RESIN	6	2	2	RA5500-211-0	61	YA9103-169-0
PS-17	BACK UP RING		SPECIAL RESIN	6	2	2	RA5500-211-0	62	YA9103-098-0
PS-18	BACK UP RING		SPECIAL RESIN	3	1	1	RA5500-211-0	63	YA9103-168-0
MFR'S NAME & ADDRESS		YOOWON INDUSTRIES LTD.							

SPARE							PAGE		
SERVO CONTROLLER FOR							SHIP No.		
6V-FH2B-MK-JANNEY PUMP							BOX No.		
NO	NAME OF PART	SKETCH	MATERIAL	SUPPLY PER SHIP			DRAWING		REMARKS
				WORK- ING	STANDARD SPARE	ADDITIONAL SPARE	No.	PART No.	
PS-31	O RING FOR VALVE CASE		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	136	361-031-075-1
PS-32	O RING FOR COVER		SYNTHETIC RUBBER	6	2	2	RA5500-211-0	137	361-031-110-1
PS-33	O RING FOR ADJUST SCREW		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	138	361-031-035-1
PS-34	O RING FOR PLUG		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	139	361-031-025-1
PS-35	O RING FOR SPACER		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	140	YA9107-115-0
PS-36	O RING FOR SPACER & AIR VENT PLUG		SYNTHETIC RUBBER	18	4	4	RA5500-211-0	141	361-031-014-1
PS-37	O RING FOR ROD		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	142	361-031-008-1
PS-38	O RING FOR ROD		SYNTHETIC RUBBER	3	1	1	RA5500-211-0	143	361-031-006-1
MFR'S NAME & ADDRESS		YOOWON INDUSTRIES LTD.							

SPARE							PAGE		
LINE FILTER FOR 6V-FH2B-MK J.P TYPE GC-12-4-20U-RIV							SHIP No.		
							BOX No.		
No.	NAME OF PART	SKETCH	MATERIAL	SUPPLY PER SHIP			DRAWING		REMARKS
				WORK- ING	STANDARD SPARE	ADDITIONAL SPARE	No.	PART No.	
PS-61	FILTER ELEMENT		PAPER	3	3	-	N76-S20-1018	2	P-GC-4-20U
PS-62	O RING FOR COVER		SYNTHETIC RUBBER	6	2	2	N76-S20-1018	9	361-031-100-2
	BACK UP RING FOR COVER		SPECIAL RESIN	6	2	2	N76-S20-1018	10	ENDLESS
PS-63	O RING FOR BODY		SYNTHETIC RUBBER	3	1	1	N76-S20-1018	13	
PS-64	O RING FOR PLUG		SYNTHETIC RUBBER	6	2	2	N76-S20-1018	15	361-031-040-2
	BACK UP RING FOR PLUG		SPECIAL RESIN	6	2	2	N76-S20-1018	16	ENDLESS
PS-65	O RING FOR INDICATOR		SYNTHETIC RUBBER	6	2	2	N76-S20-1018	18	361-031-040-2
	BACK UP RING FOR INDICATOR		SPECIAL RESIN	6	2	2	N76-S20-1018	19	ENDLESS
PS-66	O RING FOR INDICATOR		SYNTHETIC RUBBER	6	2	2	N76-S20-1018	20	
PS-67	O RING FOR INDICATOR		SYNTHETIC RUBBER	6	2	2	N76-S20-1018	22	360-031-014-1
PS-68	BACK UP RING FOR INDICATOR		SPECIAL RESIN	6	2	2	N76-S20-1018	21	ENDLESS
MFR'S NAME & ADDRESS		YOOWON INDUSTRIES LTD.							

TOOL							PAGE		
6V-FH2B-MK JANNEY PUMP							SHIP No.		
							BOX No.		
NO	NAME OF PART	SKETCH	MATERIAL	SUPPLY PER SHIP			DRAWING		REMARKS
				WORK- ING	STANDARD SPARE	ADDITIONAL SPARE	No.	PART No.	
PT1	PLIER		SPECIAL STEEL	1	—	—	525-700 -203-1		
MFR'S NAME & ADDRESS		YOOWON INDUSTRIES LTD.							

SPARE							PAGE		
INDUCTION MOTOR							SHIP No.		
							BOX No.		
No.1~No.3 : 110 kW x 6P									
NO	NAME OF PART	SKETCH	MATERIAL	SUPPLY PER SHIP			DRAWING		REMARKS
				WORK- ING	STANDARD SPARE	ADDITIONAL SPARE	No.	PART No.	
SM-1	BEARING			3	2	—			#6318C3
SM-1	BEARING			3	2	—			#7314DB
SM-3	SPARE MOTOR WITH SEAT			—	—	1			110kW x 6P Starting method: Y-Δ
SM-4	CHAIN COUPLING			—	—	1			NR8022
MFR'S NAME & ADDRESS		YOOWON INDUSTRIES LTD.							

19. PURIFIERS – MKK

(OBP SPARE)

SPARE PARTS LIST for 1 (One) YEARS NORMAL OPERATION

Model : SJ110HH (For HFO purifiers)

Symbol	Parts Name	Working Qt'y	Parts No	For 1 year Supply Qt'y	Remarks
102	O ring	1	455871002	2	
107	O ring	1	463751002	2	
111	disc (1)	1 set	184616102	1	
114	O ring	1	A30130B	2	
116	O ring	1	465269001	2	
117	O ring	1	465275001	2	
118	main seal ring	1	463754001	1	
126	O ring	4	A11205F	16	
127	O ring	4	A42019F	16	
129	O ring	2	A42019F	8	
130	valve sheet	2	436385001	4	
131	O ring	2	A41014F	8	
133	packing	1	455888001	2	
220	O ring	1	A10048A	1	
227	O ring	2	A10048A	4	
303	O ring	1	A20680A	2	
311	O ring	1	A20058A	2	
315	O ring	2	A10045A	4	
318	O ring	1	A10015A	2	
323	O ring	1	A80227A	2	
325	gasket	2	L06004215	4	
329	gasket	1	L03802815	2	
333	O ring	1	A10028A	2	
335	O ring	1	A10028A	2	for HIDENS
505	O ring	2	A20200A	2	
511	ball bearing (1)	1	I16212J8	1	
517	O ring	1	A80228A	2	
519	O ring	1	A80228A	2	
529	bearing (2)	1	I42307J1	1	
532	ball bearing (3)	1	I27307J2	1	
602	ball bearing (1)	2	I16308J1	2	
609	oil seal	1	K1507010A	1	
610	O ring	1	A20270A	1	
621	friction clutch(60Hz)	3	362593105	3	For 60Hz
625	oil seal	1	K1507010A	1	
708	O ring	2	464856001	4	
711	packing	1	463797001	2	
104	drain nozzle	2	450553001	2	
209	gasket	1	455820001	1	
216	gasket	2	L03402610	2	
217	sight glass	1	466125001	1	
218	rubber plate	4	420834001	4	4pcs/SET
238	brake lining	1	316089001	1	
240	set screw (2)	2	E100512I1	2	
310	O ring	1	A20135A	1	
502	lock washer	1	J3012C0	1	
524	O ring	1	A10140A	1	
536	steel ball	1	J1020J0	1	
605	lock washer	1	J3008C0	1	
606	retaining ring (C)	1	D3040D2	1	
624	O ring	1	A20145A	1	
706	bolt	4	F1005010B	4	
707	tongued washer	4	H5005B2	4	

SPARE PARTS LIST for 1 (One) YEARS NORMAL OPERATION

Model : SJ60HH (For M/E LO purifiers)

Symbol	Parts Name	Working Qt'y	Parts No	For 1 year Supply Qt'y	Remarks
102	O ring	1	465265001	2	
107	O ring	1	463638002	2	
111	disc (1)	1 set	254184102	1	
114	O ring	1	A30100B	2	
116	O ring	1	465268001	2	
117	O ring	1	465273001	2	
118	main seal ring	1	463641001	1	
126	O ring	4	A10011F	16	
127	O ring	4	A42017F	16	
129	O ring	2	A42017F	8	
130	valve sheet	2	436000001	4	
131	O ring	2	A41013F	8	
133	packing	1	455808001	2	
220	O ring	1	A10032A	1	
227	O ring	2	A10034A	4	
303	O ring	1	A20565A	2	
311	O ring	1	A20045A	2	
315	O ring	2	A10032A	4	
318	O ring	1	A10015A	2	
323	O ring	1	A10044A	2	
325	gasket	2	L04403215	4	
329	gasket	1	L03802815	2	
333	O ring	1	A10028A	2	
335	O ring	1	A10026A	2	for HIDENS
505	O ring	2	A20160A	2	
511	ball bearing (1)	1	I16209J8	1	
517	O ring	1	A20040A	2	
519	O ring	1	A20040A	2	
529	bearing (2)	1	I42306J1	1	
532	ball bearing (3)	1	I27306J2	1	
602	ball bearing (1)	2	I16307J1	2	
609	oil seal	1	K1456510A	1	
610	O ring	1	A20275A	1	
621	friction clutch(60Hz)	3	362592105	3	For 60Hz
625	oil seal	1	K1456510A	1	
708	O ring	2	464855001	4	
711	packing	1	463786001	2	
104	drain nozzle	2	450553005	2	
209	gasket	1	455820001	1	
216	gasket	2	L03402610	2	
217	sight glass	1	466125001	1	
218	rubber plate	4	420695001	4	(4pcs/SET)
238	brake lining	1	363875001	1	
240	set screw (2)	2	E100512I1	2	
310	O ring	1	A20120A	1	
502	lock washer	1	J3009C0	1	
524	O ring	1	A10115A	1	
536	steel ball	1	J1020J0	1	
605	lock washer	1	J3007C0	1	
606	retaining ring (C)	1	D3035D2	1	
624	O ring	1	A20135A	1	
706	bolt	4	F1005010B	4	
707	tongued washer	4	H5005B2	4	

SPARE PARTS LIST for 1 (One) YEARS NORMAL OPERATION
Model : SJ25HH (For AUX LO purifiers)

Symbol	Parts Name	Working Qt'y	Parts No	For 1 year	Remarks
				Supply Qt'y	
102	O ring	1	465264001	2	
107	O ring	1	463596002	2	
111	disc (1)	1 set	254180102	1	
114	O ring	1	A30070B	2	
116	O ring	1	465267001	2	
117	O ring	1	465271001	2	
118	main seal ring	1	463598001	1	
126	O ring	4	A10011F	16	
127	O ring	4	A42017F	16	
129	O ring	2	A42017F	8	
130	valve sheet	2	436000001	4	
131	O ring	2	A41013F	8	
133	packing	1	455812001	2	
220	O ring	1	A10032A	1	
227	O ring	2	A10028A	4	
303	O ring	1	A20425A	2	
311	O ring	1	A10034A	2	
315	O ring	2	A10024A	4	
318	O ring	1	A10016A	2	
323	O ring	1	A10031A	2	
325	gasket	2	L03802815	4	
329	gasket	1	L03802815	2	
333	O ring	1	A10028A	2	
335	O ring	1	A10026A	2	for HIDENS
505	O ring	2	A20130A	2	
511	ball bearing (1)	1	I16207J8	1	
517	O ring	1	A20030A	2	
519	O ring	1	A20030A	2	
529	bearing (2)	1	I42304J1	1	
532	ball bearing (3)	1	I27304J2	1	
602	ball bearing (1)	2	I16306J1	2	
609	oil seal	1	K1406211A	1	
610	O ring	1	A20275A	1	
621	friction clutch(60Hz)	3	362591105	3	For 60Hz
625	oil seal	1	K1406211A	1	
708	O ring	2	464854001	4	
711	packing	1	463626001	2	
104	drain nozzle	2	450553005	2	
209	gasket	1	455820001	1	
216	gasket	2	L03402610	2	
217	sight glass	1	466125001	1	
218	rubber plate	4	420695001	4	4pcs/SET
238	brake lining	1	363874001	1	
240	set screw (2)	2	E100512I1	2	
310	O ring	1	A20100A	1	
502	lock washer	1	J3007C0	1	
524	O ring	1	A10095A	1	
536	steel ball	1	J1016J0	1	
605	lock washer	1	J3006C0	1	
606	retaining ring (C)	1	D3030D2	1	
624	O ring	1	A20130A	1	
706	bolt	4	F1004008B	4	
707	tongued washer	4	H5004B2	4	

20. SEWAGE TREATMENT PLANT – NANJING LUZHOU

(OBP SPARE)



中船绿洲环保（南京）有限公司

CSSC LUZHOU ENVIRONMENT PROTECTION (NANJING) Co., Ltd.

南京市江宁滨江开发区丽水大街

Lishui Street, Binjiang Economic Development Zone, Jiangning Distric, NanJing China

TEL: +86 025 86721480 FAX: +86 025 86721689 E-mail: luzhou@nj-lzep.com Website: www.nj-lzep.com

LETTER OF OWNER'S BENEFIT

Date: 9th October, 2024

Pacific International Lines

Sewage Treatment Plant for 5 newbuildings– 13000TEU in Hudong Shipyard

HullsNo.:H1933A/H1934A/H1935A/H1936A/H1937A

Dear Sir,

We attach great importance to your building 13000TEU in China.

We are pleased to note that you are aware of our high-quality marine products. We hope to be selected by you as the maker and your partner for this new-building project.

We would request your support for our products, as listed below.

➤ STP

We are pleased to confirm the following which will be offered to PIL, (hereinafter calls OWNER) as final ship owner's benefits.

(1)10% discount for future spare parts procurement.

(2)We will provide additional one year warranty for sewage treatment plant and vacuum system.

(3)We will provide additional one year spare parts for sewage treatment plant as Appendix 1.

(4)We will provide one level swith for each sewage treatment plant as addition spare parts.

(5)Upgrade all sewage treatment plant to two air blowers and two discharge pumps.

We look forward to be of service to you and in working with you and your staff for new projects.

Thank you for your kind attention and valuable support.

Yours faithfully.

张睿 (Zhang Rui)

船用事业部部长/Director of Marine Business Department

中船绿洲环保(南京)有限公司

CSSC LUZHOU ENVIRONMENT PROTECTION(NANJING) Co., Ltd.



Appendix 1

Spares and Tools Lists

1. 空气泵备件

Spare Parts of Air Pump

序号	名称/Name	型号/Type	数量/Qty.
1	轴承		1 套
	Bearing		1 set

2. 排放泵备件

Spare Parts of Discharge Pump

序号	名称/Name	型号/Type	数量/Qty.
1	O 型密封圈	32×3.5 GB1235-76	1 只
	O-Seal		1pc
2	O 型密封圈	130×3.1 GB1235-76	1 只
	O-Seal		1pc
3	机械密封	MN108-40 GB6556-86	1 只
	Mechanical seal		1pc
4	电机轴承	6205	1 只
	Ball bearing		1pc

3. 控制箱备件

Spare Parts of El. Control Panel

中融环境科技（南京）有限公司

第 5 页 共 5 页

张

STD 系列餐厨生活污水处理装置

技术协议

序号	名称/Name	型号/Type	数量/Qty.
1	交流接触器	LC1-D09M7C	1 只
	AC Contactor		1pc
2	中间继电器	CR-MX230AC4L+ CR-M4SFB+CR-MH	1 只
	Mid-Relay		1pc
3	指示灯	AD16-22D/G31 TH	1 只
	Indicator		1pc
4	指示灯	AD16-22D/W31 TH	1 只
	Indicator		1pc
5	蜂鸣器	AD16-22SM/R31 TH	1 只
	Buzzer		1pc
6	指示灯	AD16-22D/R31 TH	1 只
	Indicator		1pc

21. CO2 FIRE FIGHTING – LINGJACK

(OBP SPARE)

3 YEAR SPARE PARTS LIST FOR CO2 SYSTEM

CO2 SYSTEM

S/N	Component name	Description	Unit	Qty.
1	Torque wrench 扭力扳手	0~30KG.M	PCS	1
2	Nitrogen gas fitting 氮气瓶充装附件		PCS	1
3	Seal ring 密封圈		PCS	20
4	Combination gasket 组合垫片	DN12	PCS	10
5	O-ring O型圈	ø 75×5.7	PCS	10
6	Start line compression nut 启动管压紧螺母		PCS	10
7	Special wrench 特种扳手	17×19	PCS	1
8	Special wrench 特种扳手	27×30	PCS	1
9	Sleeve 套筒	17	PCS	1
10	PRESSURE GAUGE 压力表及阀	SS304,0-5 Mpa	PCS	1
11	HIGH PRESSURE HOSE 高压软管	L=460mm,Ø13, Female M22	PCS	10
12	HIGH PRESSURE HOSE 高压软管	L=460mm,Ø6, Female M14	PCS	10
13	A TYPE CHECK VALVE A型止回阀		PCS	10
14	B TYPE CHECK VALVE B型止回阀		PCS	5
15	Electric spare package Fuse(1 shipset) Button (1PC.)Relay (2PCS.) Lamp (1PC.)		PCS	1
16	REPEATER(wheel house) 复示器	DC24V Munsell 7.5 BG	PCS	1

Note: S/N 16, 1 X REPEATER (wheel house), is supplied as Loose Item and not as a spare. It is to be installed by Shipyard during New Building stage

GOOD QUALITY • GREAT SERVICE

WWW.LINGJACK.COM

林杰技术发展（新加坡）有限公司

Lingjack Technology Development Pte Ltd

上海办：上海市浦东新区康杉路 151 弄康馨花园 28 号

Headquarter: No. 1 Woodlands Terrace, Lingjack Industrial

电话：(86 21) 20960301

Building, Singapore 738471

邮箱：info@ljsafe.com

Tel.: (65) 6759 4163



22. DRY CHEMICAL POWDER SYSTEM – LINGJACK

(OBP SPARE)

3 YEAR SPARE PARTS LIST FOR DRY CHEMICAL POWDER SYSTEM

DRY CHEMICAL POWDER SYSTEM

S/N	Component name	Description	Unit	Qty.
1	Torque wrench 扭力扳手		PCS	2
2	HIGH PRESSURE HOSE 释放软管	L=460mm,Ø13	PCS	1
3	SAFETY DISC OF 68L CYLINDER 68L 气瓶安全阀片		PCS	2
4	SPARE PARTS BOX 备件箱		PCS	1
5	CYLINDER FRAME For Test 测试用干粉		KG	100

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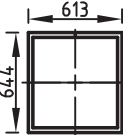
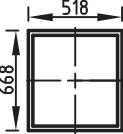
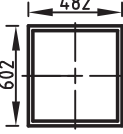
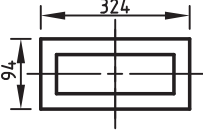
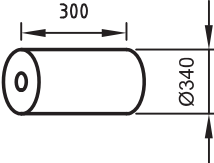
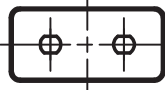
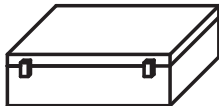
邮箱：info@ljsafe.com

Tel.: (65) 6759 4163

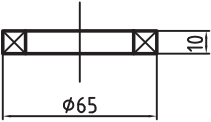
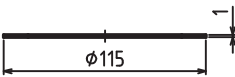


23. OILY WATER SEPARATOR – HEISHIN

(STANDARD + OBP SPARE)

SPARES		TYPE HCS-1000	B	DR. NO. 116969		PAGE		
15PPM BILGE SEPARATOR (1)						SHIP NO.		H1933A/1934A/1935A/ H1936A/1937A
						BOX NO.		
NO	NAME	SKETCH	MATERIAL	SUPPLY		DRAWING		REMARK
				WORKING PER OWS	SPARE PER SHIP	NO	PART NO	
1	Front Flange Packing	T = 5 1.3kg 	NBR	2	4			1st Stage 2nd Stage
2	Front Flange Packing	T = 5 0.9kg 	NBR	1	2			3rd Stage
3	Front Flange Packing	T = 5 0.5kg 	NBR	1	2			4th Stage
4	Front Flange Packing	T = 5 0.3kg 	NBR	2	4			2nd Stage 3rd Stage
5	Filler		HFS-EB	1 (150kg)	30kg (2pcs)			2nd Stage
6	Coalescer	11.0kg 	-	2	2			4th Stage
7	Zinc Anode	1.4kg 	Zinc	2	2			1st Stage 3rd Stage
8	Spare Parts Box		Steel	3				
9								
10								
MFR'S NAME & ADDRESS		HEISHIN PUMP WORKS CO., LTD. 5-30, 1-Chome, Furuta, Harima-cho, Kako-gun, Hyogo-pref., Japan Tel: 0794-36-3001 Fax: 0794-36-3016						

TMP NO. : HRI40D01-2Y

SPARE PARTS LIST (PUMP)						PAGE		1C-1	
PUMP NAME		BILGE SEPARATOR SERVICE PUMP				SHIP NO.		H1933A/34A/ 35A/36A/37A	
						MODEL		HRI-40D100	
NO.	NAME WEIGHT kg	SKETCH	MATERIAL	SUPPLY		DRAWING		REMARKS	
				WORKING PER PUMP	SPARE PER SHIP	NO.	PART NO.		
1	PACKING RING		RAMIE FIBER with PTFE	1 SET	1 SET	HRI-40D100100D	301		
2	CASING GASKET			1	1	HRI-40D100100D	501		
3	RELIEF VALVE SPRING		STAINLESS STEEL	1	1				
MFR'S NAME & ADDRESS		 HEISHIN PUMP WORKS CO.,LTD. 1-5-30,Furuta,Harima-cho,Kako-gun,Hyogo,Japan TEL:+81 79-436-3001 FAX:+81 79-436-3025							

24. DF BOILER + EXHAUST GAS BOILER – KANGRIM

(STANDARD+ OBP SPARE)



KANGRIM HEAVY INDUSTRIES CO., LTD.



Head office & Changwon factory

76, Yeondeok-ro, Seongsan-gu, Changwon-si, Korea

Tel : +82-55-269-7700(rep), E-mail : business4@kangrim.com

Pohang factory

1111, Hae-an-ro, Heunghae-eup, Buk-gu, Pohang-si, Korea

Tel : +82-54-230-8500, Home : www.kangrim.com

STANDARD SPARE PART LIST

Owner : PIL
Project : 13,000TEU CNTR
Shipyard : HUDONG
Hull no. : H1933A~1937A
Product : Aux. Boiler

Rev.00 (Updated : 2025.07.29)



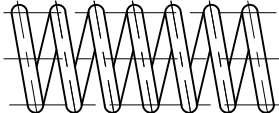
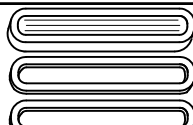
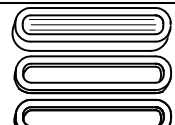
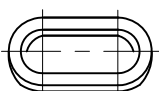
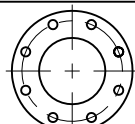
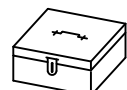
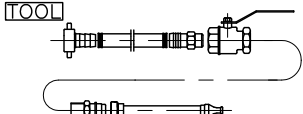
Kangrim Changwon Factory (Head office)

76, Yeondeok-ro, Sungsan-gu, Changwon city, Korea

Tel : +82-55-269-7700(ref), Home : www.kangrim.com



HULL NO.: H1933A~37A

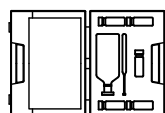
NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
		SIZE	CODE NO.	YARD WORK'G	SPARE	WEIGHT
1	SPRING		65A x 100A / SUP9	—	2	FOR SAFETY VALVE
		180 mm	2006220			1.54 kg/ea
2	GLASS & GASKET		BOROSILICATE & NON-ASBESTOS	—	1	C to C = 1200
		400 mm	—			0.52 kg/ea
3	GLASS & GASKET		BOROSILICATE & NON-ASBESTOS	—	2	C to C = 1200
		280 mm	—			0.04 kg/ea
4	GASKET		NON-ASBESTOS	1	1	FOR MANHOLE
		3.0t x ID430 x 330 mm	2001518			0.2 kg/ea
5	GASKET		SOFF / NON-ASBESTOS	8	8	FOR HANDHOLE
		10K-100A	2004830			0.05 kg/ea
6	PRESSURE GAUGE		0~20 MPa B100ø, PF3/8"	—	1	—
		B100ø	20B2340			1 kg/ea
7	SPARE BOX		—	—	1	—
		400x400x300 mm	5024778			8.41 kg/ea
8	WATER WASHING HOSE		M42 X P2 HOSE L=15m	—	1	BALL V/V 10K-25A
			4000397			3.08 kg/ea
DWG No.		SPARE OF AUX. BOILER			1 OF 2	
MB-PL-01					CODE	—

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.



HULL NO.: H1933A~37A

NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
		SIZE	CODE NO.	YARD WORK'G	SPARE	WEIGHT
9	WATER TEST KIT	TOOL 	SPECTRAPAK 310	—	1	—
			3001343			1.22 kg/ea
DWG No.		SPARE OF AUX. BOILER			2 OF 2	
MB-PL-01					CODE	—

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.



HULL NO.: H1933A~37A

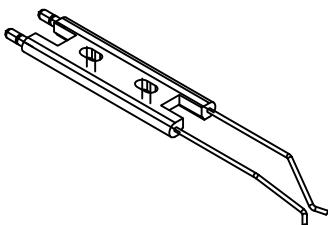
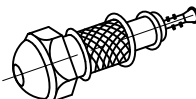
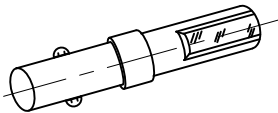
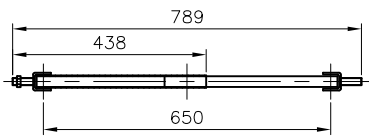
NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
			CODE NO.	YARD WORK'G	SPARE	
1	STRAINER		—	—	1	—
2	GASKET		1GPB#025 M1	—	1	—
3	MECHANICAL SEAL		1GRS#016035	—	1	—
4	BEARING		SKF6302-2RS1	—	1	—
5	O-RING		AS 2842-028 / AS 2842-033	—	1/1	—
DWG No.		SPARE OF F.O. PUMP			1 OF 1	
BNR-SP-01					CODE	—

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.



HULL NO.: H1933A~37A

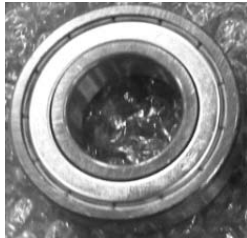
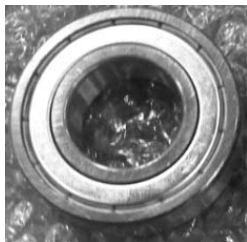
NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
			CODE NO.	YARD WORK'G	SPARE	
1	IGNITION ELECTRODES (RIGHT/LEFT)		—	—	1	BURNER
2	OIL NOZZLE		—	—	1	BURNER
3	PHOTO CELL (RAR9)		—	—	1	BURNER
4	LOCKING DEVICE		—	—	1	BURNER
DWG No.		SPARE OF BURNER			1 OF 1	
BNR-SP-01					CODE	—

* NOTE : The drawing can be different with actual product.

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HULL NO.: H1933A~37A

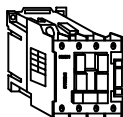
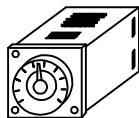
NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
			CODE NO.	YARD WORK'G	SPARE	
1	MECHANICAL SEAL		RMG12	—	1	—
2	BEARING(DE)		6308—ZZ/C3	—	1	—
3	BEARING(NDE)		6308—ZZ/C3	—	1	—
DWG No.		SPARE OF FEED WATER PUMP			1 OF 1	
MB—SP—01					CODE	—

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.



HULL NO.: H1933A/34A/35A/36A/37A

NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
		SIZE	CODE NO.	YARD WORK'G	SPARE	WEIGHT
1	CONTACTOR		9A	—	1	220V AC
		—	6009359			0.5 kg / ea
2	TIMER		H3BA	—	1	220V AC
		—	6001605			100 g / ea
3	RELAY		2N	—	1	24V DC
		—	6011550			35 g / ea
4	FUSE		1A	—	2	—
		—	6000695			22 g / ea
5						
6						
7						
8						
DWG No.		SPARE OF AUX. BOILER CONTROL PANEL			1 OF 1	
MB-PL-01					CODE	—

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.



KANGRIM HEAVY INDUSTRIES CO., LTD.



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Pohang factory

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ADDITIONAL SPARE PART LIST

Owner : PIL
Project : 13,000TEU CNTR
Shipyard : HUDONG
Hull no. : H1933A~1937A
Product : Aux. Boiler

Rev.02 (Updated : 2024.11.01)



Kangrim Changwon Factory (Head office)

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ADDITIONAL SPARE PARTS LIST Aux. Boiler			Project	13,000TEU CNTR
			Shipyard	HUDONG
			Hull no.	H1933A~1937A
			Page	1 of 3
No	Description	Sketch	Q'ty	Remark
1	Spring		2	Safety v/v
2	Glass		1	Level gauge
3	Gasket		2	Level gauge
4	Gasket		2	Manhole
5	Gasket		6	Handhole
6	Press.gauge		1	Boiler
7	Ignition electorde (Right/left)		1	Burner



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ADDITIONAL SPARE PARTS LIST Aux. Boiler			Project	13,000TEU CNTR
			Shipyard	HUDONG
			Hull no.	H1933A~1937A
			Page	2 of 3
No	Description	Sketch	Q'ty	Remark
8	Atomizer		1	Burner
9	Flame eye sensor		1	Burner
10	Igniton Transformer		2	Burner
11	Mag contactor		1	Control panel
12	Timer		1	Control panel
13	Timer		1	Control panel
14	Idec relay		1	Control panel



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ADDITIONAL SPARE PARTS LIST Aux. Boiler			Project	13,000TEU CNTR
			Shipyard	HUDONG
			Hull no.	H1933A~1937A
			Page	3 of 3
No	Description	Sketch	Q'ty	Remark
15	Idec aux relay		1	Control panel
16	Idec relay			Control panel
17	Bulb		5	Control panel
18	Fuse		2	Control panel
19	Spare box		1	



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Pohang factory

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Tel : +82-54-230-8500, Home : www.kangrim.com

STANDARD SPARE PART LIST

Owner : PIL
Project : 13,000TEU CNTR
Shipyard : HUDONG
Hull no. : H1933A~1937A
Product : Exhaust Gas Boiler (Smoke Tube)

Rev.00 (Updated : 2025.07.29)



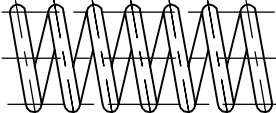
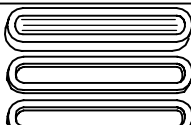
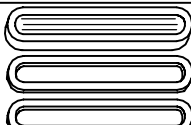
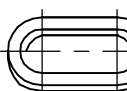
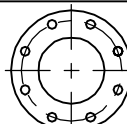
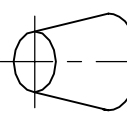
Kangrim Changwon Factory (Head office)

76, Yeondeok-ro, Sungsan-gu, Changwon city, Korea

Tel : +82-55-269-7700(ref), Home : www.kangrim.com



HULL NO.: H1933A~37A

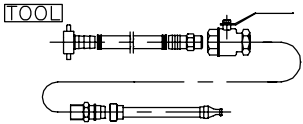
NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
		SIZE	CODE NO.	YARD WORK'G	SPARE	WEIGHT
1	SPRING		65A x 100A / SUP9	—	2	FOR SAFETY VALVE
		180 mm	2006220			1.54 kg
2	GLASS		BOROSILICATE & NON-ASBESTOS	—	1	C to C = 800 LENGTH : 370
		370 mm	2095050			0.52 kg
3	GASKET		BOROSILICATE & NON-ASBESTOS	—	1	C to C = 800 LENGTH : 220
		220 mm	3003592			0.31 kg
4	GASKET		NON-ASBESTOS	2	2	FOR MANHOLE
		3.0t x ID430 x 330 mm	2001518			0.2 kg
5	GASKET		SOFF / NON-ASBESTOS	12	12	FOR HANDHOLE
		10K-100A	2004830			0.05 kg
6	PRESSURE GAUGE		0~20 bar B100ø, PF3/8"	—	1	—
		B100ø	20B2340			— kg
7	TUBE PLUG FOR BOILER TUBE		CARBON-STEEL	—	40	—
		ø50.8	4000389			0.60 kg
8	SPARE BOX		—	—	1	—
		600x400x300 mm	5024779			11.14 kg
DWG No.		SPARE OF EXH. GAS BOILER			1 OF 2	
MB-PL-01					CODE	—

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.



HULL NO.: H1933A~37A

NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
		SIZE	CODE NO.	YARD WORK'G	SPARE	WEIGHT
9	WATER WASHING HOSE		M42 X P2 HOSE L=20m	—	1	BALL V/V 10K–25A
			4067049			3.08 kg
DWG No.		SPARE OF EXH. GAS BOILER			2 OF 2	
MB-PL-01					CODE	—

* NOTE : The drawing can be different with actual product.

KANGRIM HEAVY INDUSTRIES CO., LTD.



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Head office & Changwon factory

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Pohang factory

1111, Haean-ro, Heunghae-eup, Buk-gu, Pohang-si, Korea

Tel : +82-54-230-8500, Home : www.kangrim.com

ADDITIONAL SPARE PART LIST

Owner : PIL
Project : 13,000TEU CNTR
Shipyard : HUDONG
Hull no. : H1933A~1937A
Product : Exhaust Gas Boiler (Smoke Tube)

Rev.00 (Updated : 2024.08.27)



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Tel : +82-54-230-8500, Home : www.kangrim.com

ADDITIONAL SPARE PARTS LIST Exhaust Gas Boiler (Smoke Tube)			Project	13,000TEU CNTR
			Shipyard	HUDONG
			Hull no.	H1933A~1937A
			Page	1 of 1
No	Description	Sketch	Q'ty	Remark
1	Spring		2	Safety V/V
2	Gasket		2	For Manhole
3	Gasket		5	For Handhole
4	Press.Gauge		1	Boiler
5	Spare box		1	



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Tel : +82-54-230-8500, Home : www.kangrim.com

STANDARD SPARE PART LIST

Owner : PIL
Project : 13,000TEU CNTR
Shipyard : HUDONG
Hull no. : H1933A~1937A
Product : Exhaust Gas Economizer (Smoke Tube)

Rev.00 (Updated : 2025.07.29)



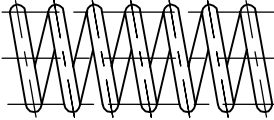
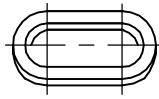
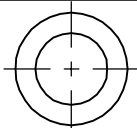
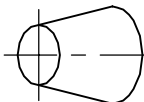
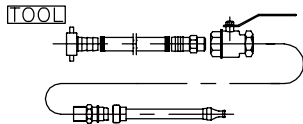
Kangrim Changwon Factory (Head office)

76, Yeondeok-ro, Sungsan-gu, Changwon city, Korea

Tel : +82-55-269-7700(ref), Home : www.kangrim.com



HULL NO.: H1933A~37A

NO.	DESCRIPTION	SKETCH	SPEC./MATERIAL	SUPPLY PER SHIP		REMARK
		SIZE	CODE NO.	YARD WORK'G	SPARE	WEIGHT
1	SPRING		40A x 65A / SUP9	—	2	FOR SAFETY VALVE
		115 mm	2006218			
2	GASKET		NON-ASBESTOS	4	2	FOR MANHOLE
		3.0t x ID430 x 330 mm	2001518			0.2 kg/ea
3	GASKET		SORF / SPIRAL WOUND	8	4	FOR HANDHOLE
		20K-100A	2001567			0.05 kg/ea
4	PRESSURE GAUGE		0~20 bar B100ø, PF3/8"	—	1	—
		B100ø	20B2340			1 kg/ea
5	TUBE PLUG FOR TUBE		CARBON-STEEL	—	40	—
		ø76.2	4001273			1.66 kg
6	SPARE BOX		—	—	1	—
		600x400x300 mm	5024779			11.14 kg/ea
7	WATER WASHING HOSE		M42 X P2 HOSE L=15m	—	1	BALL V/V 10K-25A
			4000397			3.08 kg/ea
DWG No.		SPARE OF E.G.E			1 OF 1	
MB-PL-01					CODE	—

* NOTE : The drawing can be different with actual product. **KANGRIM HEAVY INDUSTRIES CO., LTD.**



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Tel : +82-54-230-8500, Home : www.kangrim.com

ADDITIONAL SPARE PART LIST

Owner : PIL
Project : 13,000TEU CNTR
Shipyard : HUDONG
Hull no. : H1933A~1937A
Product : Exhaust Gas Economizer (Smoke Tube)

Rev.00 (Updated : 2024.08.27)



Kangrim Changwon Factory (Head office)

76, Yeondeok-ro, Sungsan-gu, Changwon city, Korea

Tel : +82-55-269-7700(ref), Home : www.kangrim.com



Head office & Changwon factory

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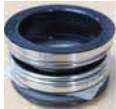
ADDITIONAL SPARE PARTS LIST Exhaust Gas Economizer (Smoke Tube)		Project	13,000TEU CNTR		
		Shipyard	HUDONG		
		Hull no.	H1933A~1937A		
		Page	1 of 1		
No	Description	Sketch	Q'ty	Remark	
1	Spring		2	Safety V/V	
2	Gasket		2	For Manhole	
3	Gasket		5	For Handhole	
4	Press.Gauge		1	Boiler	
5	Spare box		1		

25. ANTI – HEELING SYSTEM – TANKTECH

(OBP SPARE)

● Anti heeling system – Addition spare part price list

1) Spare part price list

NO.	COMPONENT	ITEM		Q'TY	WEIGHT (Approx.g)	Unit Price
800	SPARE PARTS					
801	Pump	Mechanical seal (RO(DIN)060)		1	500	\$600
802	Motor	Bearing (DE:6316C4 / N-DE:6316C4)		2	700	\$350
811	Panel	Indication lamp cap (Green & Red)		2	2	\$5
812		Indication lamp (KH-2203F AC220- White)		1	88	\$50
813		Push button (Ø22) (KH-2201-11)		1	42	\$50
814		Relay (HR705-4PL)		2	35	\$35
815		Glass Fuse (250V 2A)		2	0.6	\$35
816		Fuse (SB-C1 500V 3A)		2	22	\$50
817		Select switch (KSL25S3)		1	46	\$50
818		Indication lamp (Ø8-LED/Green & Red)		2	8	\$3

26. LEVEL GAUGING SYSTEM – HANLA IMS

(OBP SPARE)



PAGE	1 / 1
PROJECT NAME	Hudong Zhounghua - H1933A/34A/35A/36A/37A
DWG No.	HLS-REF-OBP-0001

[illegible]

27. VALVES REMOTE CONTROL SYSTEM – HANLA IMS

(OBP SPARE)



TITLE

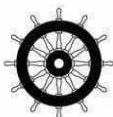
SPARE PART LIST FOR O.B.P

PAGE	1 / 1
PROJECT NAME	Hudong Zhonghua H1933A/34A/35A/36A/37A
DWG No.	HLS-CHZ2087-OBP-MAIN-0001

NO.	PART NO.	DESCRIPTION		MAKER	SPARE Q'TY	UNIT PRICE	TOTAL	REMARK
		SPECIFICATION	SKETCH					
1 - 1	VR02577	Limit switch box (Non-Ex. O/S Type)		HKC	2	\$ 290	\$ 580	APL-910N
1 - 2	VR02574	Limit switch box (Non-Ex. Throttle Type)		HKC	1	\$ 550	\$ 550	APL-905N
1 - 3	VR02683	Limit switch box (Ex-d IIC T6. O/S Type)		HKC	1	\$ 600	\$ 600	APL-510N
1 - 4	VR01380	R-I Convertor (M5MS-A-R)		M-SYSTEM	1	\$ 690	\$ 690	for Throttle type
1 - 5	VR02787	Emergency hand pump Connector		HANLA IMS	4	\$ 110	\$ 440	For Actuator
2 - 1	VR01130	Magnetic Contact (HGC 40 22NS A220)		LS	1	\$ 320	\$ 320	for Local control panel
2 - 2	VR01498	Circuit Breaker 10A		LS	1	\$ 100	\$ 100	for Local control panel
3 - 1	VR00508	Seal kit for HLT-35	-	HANLA IMS	2	\$ 140	\$ 280	for EHLT-35 /EX-EHLT-35
3 - 2	VR00509	Seal kit for HLT-70	-	HANLA IMS	2	\$ 150	\$ 300	for EHLT-70
3 - 3	VR00510	Seal kit for HLT-135	-	HANLA IMS	1	\$ 180	\$ 180	for EHLT-135
3 - 4	VR00643	Seal kit for HLTS-35	-	HANLA IMS	1	\$ 160	\$ 160	for EHLTS-35
3 - 5	VR00644	Seal kit for HLTS-70	-	HANLA IMS	1	\$ 170	\$ 170	for EHLTS-70
						AMOUNT	\$ 4,370	

28. INCINERATOR – TEAMTEC

(OBP SPARE)



Appendix A

GS900 addition spare parts

Item	Description	Code	Qty	Remark
1	Photo resistor	QRB1	1pc	For burner
2	Stator of pump		1pc	For dosing pump
3	O-seal ring	20x2.6	1pc	For circle pump
4	O-seal ring	130x3.1	1pc	For circle pump
5	Bearing	6312	1pc	For fan motor
6	Bearing	6212	1pc	For fan motor
7	Bearing	6203	1pc	For dosing pump motor
8	Bearing	6204	1pc	For dosing pump motor
9	Bearing	6202	1pc	For burner motor
10	Bearing	6205	1pc	For circ. pump motor

29. FRESH WATER GENERATOR – ALFA LAVAL

(OBP SPARE)

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Fresh Water Generator - OBP Spare List

Part Number	Quantity per Shipset	Description
AB-SP-ONE	1	1 year Spare parts package
985 25692-22	1	1y Spares - Coil for Solenoid valve
984 10230-08	1	1y Spares - Mechanical seals for PVVF-1532
984 10230-03	1	1y Spares - Chamber gaskets for PVVF-1532
984 10230-06	1	1y Spares - Seal ring(s) for PVVF-1532
984 20653-00	1	1y Spares - Drive end 6204-ZZ bearing for 80
984 20645-00	1	1y Spares - Non-Drive end 6203-ZZ bearing for 80
985 21472-14	2	1y Spares - Diaphragms for DV & FVF valves
985 00087-18	1	1y Spares - EJ pump Chamber gasket
985 00087-11	1	1y Spares - EJ pump Lower wear ring
985 00087-10	1	1y Spares - EJ pump Upper wear ring
985 00087-14	1	1y Spares - EJ pump Mech. Seal
985 00085-37	1	1y Spares - Drive end 6308-ZZ bearing for 132
985 00095-75	1	1y Spares - Non-Drive end 6207-ZZ bearing for 132

30. ME & AE FO SUPPLY UNIT – AURAMARINE

(OBP SPARE)

Qty	Item name	M/E FSU	ID code	GE FSU	ID code
1	Shaft seal for F.O. supply pumps	ACG045K7 NTBP DN50	AFS000839	ACE038K3_NTBP, DN40	AFS000801
1	Shaft seal for F.O. circulation pumps	ACG052N7_NTBP, DN65	AFS000840	ACG060K7_NTBP, DN80	AFS000984
1	Set of gaskets for automatic filter	6.64.1, 10um, DN100, 230V	AFS000836	6.72.1, 34um, DN65, 230V	AFS000796
1	Pressure switch	P004	CM020394	P004	CM020394
1	Safety valve for heaters	V027	CM003713	V027	CM003713
1	Viscometer sensor, can be used either for		AFS000972		
	ME or GE's FU				

31. FUEL AND LUB OIL COUNTER (FLOW METER) – E+H

(STANDARD + OBP SPARE)

Endress+Hauser People for Process Automation				3 years operation spare parts		Document No. 2024-13589EHC3	
Project:		HUDONG JUNGHWA SHIPYARD		Equipment Description:		Revision	
Project No./Hull ID.		H1933A-37A		Mass Flow Meter		1	
Builder:				Supplier:		Issue Date	
Owner:		PIL		Supplier Order No.:		2024/11/19	
				ENDRESS+HAUSER			

Product Name	Product Code	Quantity
I/O Module for Flow meter 485	XPD0037-AAF100	1
Power Module	71323007	1

3

32. LOCAL WATER MIST SYSTEM – JCI SEAPLUS

(STANDARD + OBP SPARE)

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Water Based Local Fire Fighting System - Standard Spare List

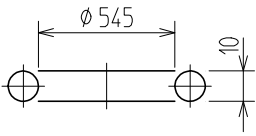
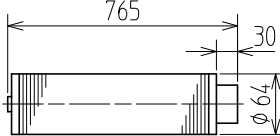
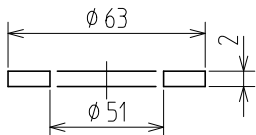
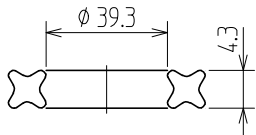
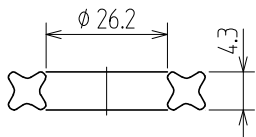
Q'ty	Code	Detail
1	MA000591	SPARE PART BOX, 200x200x100,SS400(OB)
1	MA000796	RELAY 4a-4b, DC 24V(OB)
2	MA000723	FUSE, 2A(OB)
2	PO000406	FUSE, 4A(OB)
1	MA000770	PILOT LAMP, ⌀ 22, DC 24V, WHITE(OB)
1	PM000688	PILOT LAMP WITH PUSH BUTTON, DC 24V, GREEN(OB)
1	CS001035	SPARE PILOT LAMP, ⌀ 22, DC 24V, ORANGE(OB)
1	MA000769	PILOT LAMP, ⌀ 22, DC 24V, RED(OB)
6	PM004988	LOCAL WATER MIST NOZZLE WITH PROTECTION COVER, SPF-3, MED©(OB)
1	DACOS20A	SECTION VALVE WITH COILS, 20A, BRASS, 24V DC, PG11, PU225-06A (DACOS)(OB)
1	DACOS32A	SECTION VALVE WITH COILS, 32A, BRASS, 24V DC, PG11, PU225-12A (DACOS)(OB)
1	DACOS40A	SECTION VALVE WITH COILS, 40A, BRASS, 24V DC, PG11, PU225-14A (DACOS)(OB)

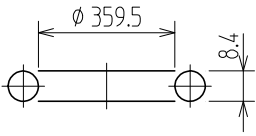
PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Water Based Local Fire Fighting System - OBP Spare List

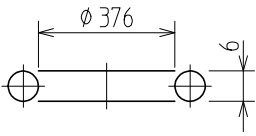
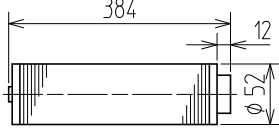
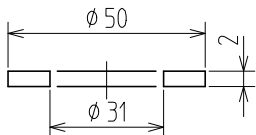
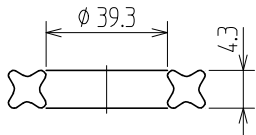
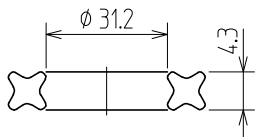
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1	MA000796	RELAY 4a-4b, DC 24V(OB)
2	MA000723	FUSE, 2A(OB)
2	PO000406	FUSE, 4A(OB)
1	MA000770	PILOT LAMP, ⌀ 22, DC 24V, WHITE(OB)
1	PM000688	PILOT LAMP WITH PUSH BUTTON, DC 24V, GREEN(OB)
1	CS001035	SPARE PILOT LAMP, ⌀ 22, DC 24V, ORANGE(OB)
1	MA000769	PILOT LAMP, ⌀ 22, DC 24V, RED(OB)
6	PM004988	LOCAL WATER MIST NOZZLE WITH PROTECTION COVER, SPF-3, MED©(OB)
1	DACOS20A	SECTION VALVE WITH COILS, 20A, BRASS, 24V DC, PG11, PU225-06A (DACOS)(OB)
1	DACOS32A	SECTION VALVE WITH COILS, 32A, BRASS, 24V DC, PG11, PU225-12A (DACOS)(OB)
1	DACOS40A	SECTION VALVE WITH COILS, 40A, BRASS, 24V DC, PG11, PU225-14A (DACOS)(OB)

33. AUTOBACKFLUSHING FILTER – KANAGAWA

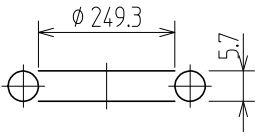
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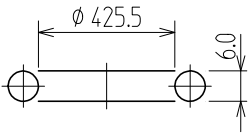
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						BOX NO.		
NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SET		DRAWING		REMARKS
				WORK-ING	SPARE	NO.	PART NO.	
1	O-リング O-RING		NBR	1X2	1		A-3	OR0545/10-BR
2	エレメント ELEMENT		SUS304 ALUMINIUM NBR	21X2	1		B-1 B-2	E7SXLE-EL-01
3	パッキン PACKING		NBR	21X2	21		B-2	P063/51/2-BRC
4	X-リング X-RING		NBR	1X2	1		D-7	R38-BR
5	X-リング X-RING		NBR	1X2	1		E-4	R25-BR (for ACTUATOR)
6								
7								
8								
9								
10								
MFR'S NAME & ADDRESS		KANAGAWA KIKI KOGYO CO., LTD. Yokohama, Japan				LIST NO.		E7SX07-LE-SP
						M.F.G. NO.		

						PAGE		
SLUDGE CHECKER SPARE PARTS LIST / スラッジ・チェッカー予備品リスト M/E L.O. AUTO. FILTER						SHIP NO.		
						BOX NO.		
NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SET		DRAWING		REMARKS
				WORK- ING	SPARE	NO.	PART NO.	
1	O-リング O-RING		NBR	1	1		H-3	P360-BR
2								
3								
4								
5								
6								
7								
8								
9								
10								
MFR'S NAME & ADDRESS		KANAGAWA KIKI KOGYO CO., LTD. Yokohama, Japan				LIST NO.		XA-53/54-SP
						M.F.G. NO.		

						PAGE		
SPARE PARTS LIST / 予備品リスト M/E TURBO CHARGER AUTO. FILTER						SHIP NO.		
						BOX NO.		
NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SET		DRAWING		REMARKS
				WORK- ING	SPARE	NO.	PART NO.	
1	O-リング O-RING		NBR	1	1		A-3	V380-BR
2	エレメント ELEMENT		SUS304 ALUMINIUM NBR	16	1		B-1 B-2	E4X-EL-03
3	パッキン PACKING		NBR	16	16		B-2	P050/31/2-BRC
4	X-リング X-RING		NBR	1	1		D-7	R38-BR
5	X-リング X-RING		NBR	1	1		E-3	R30-BR (for ACTUATOR)
6								
7								
8								
9								
10								
MFR'S NAME & ADDRESS		KANAGAWA KIKI KOGYO CO., LTD. Yokohama, Japan				LIST NO.		E4X0Z-SP-MINI
						M.F.G. NO.		

						PAGE		
SLUDGE CHECKER SPARE PARTS LIST / スラッジ・チェッカー予備品リスト M/E TURBO CHARGER AUTO. FILTER						SHIP NO.		
						BOX NO.		
NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SET		DRAWING		REMARKS
				WORK- ING	SPARE	NO.	PART NO.	
1	O-リング O-RING		NBR	1	1		H-3	P360-BR
2								
3								
4								
5								
6								
7								
8								
9								
10								
MFR'S NAME & ADDRESS		KANAGAWA KIKI KOGYO CO., LTD. Yokohama, Japan				LIST NO.		XA-53/54-SP
						M.F.G. NO.		

						PAGE		
SPARE PARTS LIST / 予備品リスト M/E TURBO CHARGER L.O. BY-PASS FILTER						SHIP NO.		
						BOX NO.		
NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SET		DRAWING		REMARKS
				WORK-ING	SPARE	NO.	PART NO.	
1	O-リング O-RING		NBR or FPM	1	1		13	G250
2								
3								
4								
5								
6								
7								
8								
9								
10								
MFR'S NAME & ADDRESS		KANAGAWA KIKI KOGYO CO., LTD. Yokohama, Japan				LIST NO.		SS-F0020A-100A-SP
						M.F.G. NO.		

						PAGE		
SPARE PARTS LIST / 予備品リスト M/E L.O. BY-PASS FILTER						SHIP NO.		
						BOX NO.		
NO.	NAME	SKETCH	MATERIAL	SUPPLY PER SET		DRAWING		REMARKS
				WORK- ING	SPARE	NO.	PART NO.	
1	O-リング O-RING		FPM	1	1		15	V430-FPM
2								
3								
4								
5								
6								
7								
8								
9								
10								
MFR'S NAME & ADDRESS		KANAGAWA KIKI KOGYO CO., LTD. Yokohama, Japan				LIST NO.		RS-D1000A-250-SP
						M.F.G. NO.		

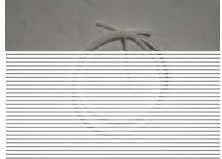
34. BALLAST WATER TREATMENT PLANT – HEADWAY

(OBP SPARE)

List of Supplied Spare Parts

No.	Item	Spec.	Picture	Qty.	Unit	ERP No.	Remark
1	O Seal Strip	Φ7_ Fluororubber		2.5	M	01.20.005.001	Filter Spare ERP No:50.51.244.007
2	O Seal Strip	Φ5.3_ Fluororubber		2.5	M	01.20.005.002	
3	Spring	76*66.5_316		3	Pcs	05.10.151.005	
4	Suction nozzle1	F-220XZ-R295-00		3	Pcs	50.12.517.010	
5	Suction nozzle2	F-220XZ-00		3	Pcs	50.12.517.003	

No.	Item	Spec.	Picture	Qty.	Unit	ERP No.	Remark
6	X-ring	Φ46.99x5.33_ Fluororubber		2	Pcs.	05.11.202.001	
7	O-ring	Φ112x3.55_ Fluororubber		1	Pcs.	05.11.102.008	
8	O-ring	Φ191x5.3_ Fluororubber		1	Pcs.	05.11.102.021	
9	O-ring	Φ138x5.3_ Fluororubber		1	Pcs.	05.11.102.018	
10	O-ring	Φ61.5x2.65_ Fluororubber		3	Pcs.	05.11.102.056	

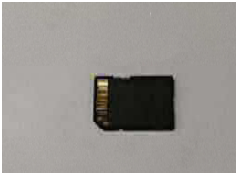
No.	Item	Spec.	Picture	Qty.	Unit	ERP No.	Remark
11	Tool For Lifting Filter Element	M10_316-N_GB/T825		2	Pcs.	05.02.506.002	
12	Plastic Pipe-Air	$\phi 8$		0.5	M	01.05.005.003	Monitoring Unit Spare ERP No:50.51.145.001
13	Plastic Pipe-Air	$\Phi 6$		0.5	M	01.05.005.001	
14	PFA Pipe	$\phi 2$		1	M	01.05.005.015	
15	Joint of Pipe	ZG1/2 to $\phi 12$ SS Pipe		1	Pcs	05.13.052.005	
16	Joint of Pipe	G1/4 to $\Phi 8$ copper pipe; Nickel plated brass		2	Pcs.	05.13.051.002	

No.	Item	Spec.	Picture	Qty.	Unit	ERP No.	Remark
17	Elbow Joint of Pipe	G1/4 to Φ8Nickel plated brass		1	Pcs.	05.13.251.001	
18	Joint of Pipe	1/4 to Φ8Nickel plated brass		2	Pcs.	05.13.051.007	
19	Relay	RIF-0-RPT-24DC/21		2	Pcs.	25.04.001.006	Filter Control Box Spare
20	Hinge	CL209Z3A001		2	Pair	05.14.001.001	Cabinet Spare
21	Cabinet Key	--		2	Pcs.	05.17.001.001	Cabinet Spare
22	Joint of Pipe	ZG1/2 to Φ 8 SS Pipe		2	Pcs.	05.13.052.003	Neutralization Unit Spare

No.	Item	Spec.	Picture	Qty.	Unit	ERP No.	Remark
23	Joint of Pipe	ZG1/2 to ϕ 12 SS Pipe		1	Pcs.	05.13.052.005	Neutralization Unit Spare
24	Single Electric Control Solenoid Valve	G1/4 Single Electric Control 24DC_SCG531C001MS (asco)		1	Pcs.	06.04.001.007	Pneumatic Valve Spare
25	Electromagnetic Coils	SCG531 (asco)		2	Pcs.	1.46.01.0016	Pneumatic Valve Spare
26	Hex head galvanized flange face screw	M8*30_GB/T5780_8.8		10	Pcs.	05.01.061.006	EUT Spare ERP No:50.51.143.007
27	Shock proof gasket	NLX8sp		6	Pcs.	05.05.403.001	
28	X-ring	Φ 34.52x3.53_ Fluororubber HMT-(50-800)		4	Pcs	05.11.202.011	
29	Sealing Gasket of Conductor Post	HMT-(50-800)		2	Pcs	50.13.305.001	

No.	Item	Spec.	Picture	Qty.	Unit	ERP No.	Remark
30	EUT cover gasket	HMT-800		3	Pcs.	50.13.302.031	
31	Filter Cup Set Of MU	--		1	Pcs.	10.01.151.022	
32	SD Card	SSK; SCRM330		1	Pcs.	1.62.04.1181	Card Spare No:50.51.147.002
33	SIMATIC S7 Card	SIMATIC S7 Card; 4 MB; 6ES7 954-8LC03-0AA0		1	Pcs.	25.12.003.020	
34	TF Card	micro SD Card; 8G		1	Pcs.	1.64.02.0027	



No.	Item	Spec.	Picture	Qty.	Unit	ERP No.	Remark
35	SD Card	SD Memory Card		2	Pcs.	25.12.001.017	

35. MAIN SWITCHBOARD, EMERGENCY
SWITCHBOARD & TRANSFORMER –
SCHNEIDER

(OBP SPARE)

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Switchboard & Transformer - Standard Spare List

NO. 编号	NAME 名称	DESCRIPTION 型号	订货号	QTY
1	Fuse, 25A 500V 100kA	RT28-32 25A		4
2	Fuse, 16A 500V 100kA	RT28-32 16A		2
3	Fuse, 10A 500V 100kA	RT28-32 10A		2
4	Fuse, 6A 500V 100kA	RT28-32 6A		6
5	Fuse, 4A 500V 100kA	RT28-32 4A		40
6	Fuse, 2A 500V 100kA	RT28-32 2A		60
7	Fuse, 1A 500V 100kA	RO15 1A		15
8	Fuse, 0.5A 500V 100kA	RO15 0.5A		10
9	Cover for pushbutton	F3-1		5
10	Cover for pushbutton	F3-3		3
11	OF Indication Contact,For NSX	29450		3
12	MX,AC220V~250V,For NSX	LV429387		3
13	Indication Lamp, AC440V,White,LED	XB4BV8B1		10
14	Indication Lamp, AC220V,White,LED	XB4BVM1		10
15	Indication Lamp, AC220V,Red,LED	XB4BVM4		6
16	Indication Lamp, AC220V,Green,LED	XB4BVM3		6
17	Indication Lamp, DC24V,Red,LED	XB4BVB4		3
18	Indication Lamp, DC24V,Green,LED	XB4BVB3		3
19	Indication Lamp, DC24V,White,LED	XB4BVB1		2
20	TIME ON DELAY,0.1~30s	LADT2		6
21	TIME ON DELAY,10~180s	LADT4		2
22	TIME OFF DELAY,1~30s	LADR2		3
23	TIME OFF DELAY,10~180s	LADR4		5
24	PUSHBUTTON BODY,1NC	ZB4BZ102		6
25	PUSHBUTTON BODY,1NO	ZB4BZ101		8
26	PUSHBUTTON BODY,1NO1NC	ZB4BZ105		5
27	PUSHBUTTON BODY,2NC	ZB4BZ104		1
28	PUSHBUTTON BODY,2NO	ZB4BZ103		1
29	PUSHBUTTON AUX CONTACT, 1NC	ZBE102		8
30	PUSHBUTTON AUX CONTACT, 1NO	ZBE101		8
31	PUSHBUTTON HEAD, GREEN	ZB4BA2		2
32	PUSHBUTTON HEAD, RED	ZB4BA4		8
33	PUSHBUTTON HEAD, BLACK	ZB4BA3		8
34	PUSHBUTTON HEAD, YELLOW	ZB4BA5		1
35	PUSHBUTTON HEAD WITH LAMP, GREEN	ZB4BW333		1
36	PUSHBUTTON HEAD WITH LAMP, RED	ZB4BW343		2
37	SELECT SWITCH HEAD WITH LAMP, ORANGE, 2 POSITION	ZB4BK1253		3
38	SELECT SWITCH HEAD, BLACK, 2 POSITION	ZB4BD2		6
39	SELECT SWITCH HEAD, BLACK, 3 POSITION	ZB4BD3		1
40	SELECT SWITCH HEAD, BLACK, 3 POSITION, SPRING	ZB4BD5		3
41	SELECT SWITCH BODY WITH KEY, BLACK, 2 POSITION	ZB4BG2		1
42	PUSHBUTTON BODY WITH LAMP, WHITE, 1NO, DC24V	ZB4BW0B11		3
43	PUSHBUTTON BODY WITH LAMP, WHITE, 2NO, AC220V	ZB4BW0M13		6
44	LAMP BODY WITH 220/6V TRANSFORMER, 60Hz	ZB4BV4		2
45	LAMP BODY WITH 440/6V TRANSFORMER, 60Hz	ZB4BV8		2
46	LAMP HEAD ,CLEAR	ZB4BV07		4
47	Aux. Contact, Side, 2NO, For Contactor	LAG8N203P		3
48	Aux. Contact, Side, 2NO, For Contactor	LAG8N203		3
49	Aux. Contact, Side, 1NO1NC, For Contactor	LAG8N113P		3
50	Aux. Contact, Side, 1NO1NC, For Contactor	LAG8N113		3
51	Aux. Contact, Front, 2NO2NC, For Contactor	LADN22		6
52	Aux. Contact, Front, 3NO1NC, For Contactor	LADN31		3
53	Aux. Contact, Front, 4NO ,For Contactor	LADN40		3
54	Aux. Contact, Front, 4NC, For Contactor	LADN04		2
55	LED SUPPRESSOR, 220V	ZBVGM1T		2
56	MIN. RELAY, AC220V, 4CO	RXM4AB2P7		20
57	MIN. RELAY, DC24V, 4CO	RXM4AB2BD		20
58	MIN. RELAY SOCKET	RXZE2M114M		20
59	MIN. RELAY AUX	RXZ400		20
60	DIODE AUX, DC6~250V	RXM040W		10
61	MIN. RELAY, DC24V, 2CO	RXG23BD		10
62	MIN. RELAY, AC220V, 2CO	RXG23M7		10
63	MIN. RELAY SOCKET	RGZE1S48M		20
64	MIN. RELAY AUX	RGZR215		20
65	RELAY, AC440V, 3NO2NC	CAD32R7		2
66	RELAY, AC220V, 3NO2NC	CAD32M7		3
67	RELAY, AC220V, 5NO	CAD50M7		2
68	RELAY, DC24V, 3NO2NC	CAD32BD		2
69	MECHA. LOCK AUX, AC220V	LAD6K10M		2
70	Fuse Carrier 10.5x38	A9N15636		20

36. GYRO COMPASS, AUTO PILOT &
MAGENETIC COMPASS – TKC

(OBP SPARE) – STORED IN MAKER
WAREHOUSE

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Gyro Compass - OBP Spares		
S/N	DESCRIPTION	QTY / SHIPSET
1	Sensitive Element for Gyro Compass	1

37. NAVIGATION & SIGNAL LIGHTS & LIGHTING FIXTURES – HAIXIN

(STANDARD + OBP SPARE)

Spare list for Shipowner

13000TEU CONTAINER VESSEL (H1933A~H1937A)

Projector/ lighting fixtures , flood light

No	Name	Type	Specification	Unit	Qty in TA	Additional Qty for Owner	Remark
1	LED tube	For JCY24-2D/AD	220V 10W	PCS	100	100	
2	LED tube	For JCY44-2D/AD	220V 20W	PCS	20	100	
3	LED驱动电源 LED DRIVE				8	8	For TG29-200 : 2只 For TG29-400 : 2只 For TG20-L : 2只 For TG18-L : 2只
4	LED bulb	For CCD9-5	220V10W E27	PCS	20	20	
5	Lampshade	For JCY24-2D/ JCY24-2AD		PCS	40	40	w/rubber sealing
6	Lampshade	For JCY44-2D/ JCY44-2AD		PCS	0	10	w/rubber sealing
7	RED CEILING LIGHT lampshade	For JPY21-2ND		PCS	2	2	
8	Lampshade	For JPY21-2ND		PCS	10	10	
9	Navigation light shade		For CXH1/2/3/4/6- 102PL	PCS	3	3	
10	Signal light shade		For CXH11-L	PCS	3	3	
11	LED DRIVE		For CXH1/2/3/4/6- 102PL	PCS	0	5	
12	LED DRIVE		For CXH11-L	PCS	0	5	

38. FIRE DETECTION SYSTEM – CONSILLIUM

(STANDARD + OBP SPARE)

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Fire Detection System - Spare List

S/N	Description	Standard Spares Quantity	OBP Spares Quantity	Sum Of Quantity
1	Optical Smoke and Heat Detector	3	5	8
2	Optical Smoke and Heat Detector Ex		1	1
3	Heat Detector High Temp	1	2	3
4	Flame Detector	1	1	2
5	Call Points (dry type)	2	2	4
6	Call Points (wet type)		3	3

39. REEFER CONTAINER SOCKET – WISKA

(OBP SPARE)

SP-VARI-PI/LED/25kA/BC/NUTS

Spare part package



22 sets of this standard spare part package will be delivered per shipset for
H1933A/H1934A/H1935A/H1936A/H1937A

Products may differ in size, colour and appearance to those shown. If you require further information or customisation, please contact us directly.

WISKA-Order no.	22000606
Type	SP-VARI-PI/LED/25kA/BC/NUTS
Sparepart	Spare part package
Material protective cap	Polyamide
Socket insert	3 - poles + PE
Clock face position	3h
Breaking capacity	25 kA
Operational areas	Outdoor
Temp. range min	-25 °C
Temp. range max	45 °C
Mounting	Wall, Surface mounting

Remarks:

Spare part package for
VARITAIN PushIn 25kA/32A containing:

1 set SP-SM/LED25kA/PI/inkl. LED:

contents:

1x Socket module with LED for
switching status and WISKA MCCB
25kA pre-wired
4x cap nuts M6

1 set SP-CAP NUTS M6/C/OT/OTE/PI:

contents:

Your Contact:

WISKA Hoppmann GmbH
Kisdorfer Weg 28 · 24568 Kaltenkirchen · Germany
☎ +49-4191-508-0 · 📠 +49-4191-508-209
contact@wiska.de · www.wiska.com

10x cap nuts (brass, nickel-plated)
for lid, socket flange and socket
module
10x lock washers

1 set SP-BC/FS-CEE/BFS/OTE/PI

contents:

1 x Bayonet cap CEE 32A
(non corrosive material)
with non corrosive fastening belt

14.02.2024

Your Contact:

WISKA Hoppmann GmbH

Kisdorfer Weg 28 · 24568 Kaltenkirchen · Germany

 +49-4191-508-0 ·  +49-4191-508-209

contact@wiska.de · www.wiska.com

40. REEFER CONTAINER MONITORING SYSTEM – NANJING FRIENDS

(STANDARD + OBP SPARE)

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Reefer Container Monitoring System - Standard Spare List

Standard Spare Part List			
1	Grasp 3.0 Installation flash drive	1 pc	

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - Reefer Container Monitoring System - OBP Spare List

OBP Spare Part List			
1	Grasp 3.0 Installation flash drive	2 pcs	
2	Power transceiver fuses	10 pcs	
3	Ink Cartridge	3 pcs	

41. AMP (INCLUDING CABLE REEL) – COSCO

(OBP SPARE)

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - AMP - OBP SPARE LIST

No.	Material	Type	Technical parameter	Brand	Qty.
1	Breaker	IC65N C20A/3P A9F18320	Rated current: 20A, 3P Trip curve: C型	Schneider	1
2	Contactor	LC1D09BD	24VDC 9A, 3P	Schneider	1
3	Pilot lamp	XB2BVB1LC	24VDC, White	Schneider	1
4	Pilot lamp	XB2BVB3LC	24VDC, Green	Schneider	1
5	Pilot lamp	XB2BVB4LC	24VDC, Red	Schneider	2
6	Button	XB2BA21C	Black, NO	Schneider	1
7	Selector switch	XB2BD21C	Short shank, NO	Schneider	1
8	Selector switch	XB2BJ53C	Long handle, NO	Schneider	1
9	Relay	MY2N-GS	24VDC 2NO/NC	Omron	2

42. CCTV – DONGXIN

(OBP SPARE)

PIL 13K TEU Hudong Shipyard New Builds (H1933A-H1937A) - CCTV - OBP Spare List

Category	Description	Qty/Ship	Unit
Wiper	Wiper for HS0110 camera	1	Pcs
Electrical Machinery	Electrical machinery for wiper	1	Pcs
Jumper	Fiber jumper wire	2	Pcs
LAN cable	2m LAN cable	2	Pcs
Camera	HS3002 camera	1	Pcs

43. BOG COMPRESSOR – DONGHWA
PNEUTEC

(OBP SPARE)

APPENDIX.1 Additional spare parts

No.	Description	Maintenance (HOUR)	Q'TY	REMARK
1	Oil filter element	4,000HOUR (1YEAR)	1+1	
2	Oil separator element	4,000HOUR (1YEAR)	1+1	
3	Gas discharge filter element(H-type)	4,000HOUR (1YEAR)	1+1	
4	Gas discharge filter element(U-type)	4,000HOUR (1YEAR)	1+1	

APPENDIX.2 Spare parts price table

No.	Description	Q'TY	Maintenance (HOUR)	Discounted price(20%) / PCS_USD	
1	Oil filter element	1EA	4,000HOUR (1YEAR)	330	264
2	Oil separator element	1EA	4,000HOUR (1YEAR)	2,120	1,696
3	Gas discharge filter element(H-type)	1EA	4,000HOUR (1YEAR)	1,190	952
4	Gas discharge filter element(U-type)	1EA	4,000HOUR (1YEAR)	1,190	952

44. LNG FUEL SUPPLY PUMP – CRYOSTAR

(OBP SPARE) – STORED IN MAKER FACTORY

the Equipment to the Shipbuilder (including but not limited to the clauses related to warranty, limitation of liability, exclusion of consequential damages etc).

Other items of the Equipment which are not subject to the Extended Warranty Period shall be guaranteed as per the mutually agreed terms and conditions of purchase related to the contract for the supply of the Equipment.

2. Kit of Operational Spare parts

One kit of the spare parts listed below will be provided to Shipowner, on delivery of the New Build Vessel Hull N° H1933.

Position	Item N°	Description	Quantity
66	798170409	O-ring	16
15	798171105	O-ring	16
46	798160497-01	Spiral wound gasket	16
14	798160683-01	Spiral wound gasket	24

3. ASSIGNMENT

Neither Party may assign this Agreement to any third party without the other Party's prior written consent.

4. VARIATION

None of the provisions of this Agreement can be amended without the written agreement of the parties.

5. LAW AND JURISDICTION

This Agreement shall be construed and take effect in accordance with English law.

The Parties shall attempt to resolve any question, dispute or difference between the Parties arising out of the Agreement amicably. In the event no settlement is reached within thirty (30) calendar days, either Party shall have the right to refer the disputed matter to arbitration. Arbitration proceedings shall be in accordance with rules and regulations of Conciliation and Arbitration of the London Maritime Arbitrator Association (LMAA). The venue of the arbitration shall be in London, United Kingdom. The arbitrators shall be three (3) in number and shall be appointed in accordance with above rules and regulations. The award of the arbitrators shall be final and binding on the Parties, and there will not be any recourse to the local courts except as may be necessary to enforce the arbitral award during any arbitration.

6. EFFECTIVENESS – CONDITIONS PRECEDENT

The Supplier's obligations as set forth herein shall be effective and enforceable by the Shipowner subject to the following conditions being fulfilled:

- A) The Shipowner shall irrevocably and unconditionally select and approve the Supplier for the supply of the Equipment in the frame of the New Build Vessel; and
- B) The Supplier shall agree upon the terms and conditions of a supply contract with the Shipbuilder for the supply of the Equipment.

7. COMPLIANCE WITH LAWS

Parties represent and warrant that they are and will be in compliance with all applicable laws (1) that prohibit corruption, bribery, cartel, money laundering, public conflict of interests, public improbity, or fraud, and (2) that regulate gifts, gratuities, and expenses paid to or on behalf of

45. LNG TANK LEVEL RADAR & TEMPERATURE MONITORING SYSTEM – EMERSON

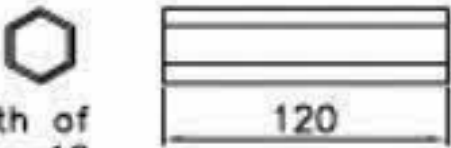
(STANDARD + OBP SPARE)

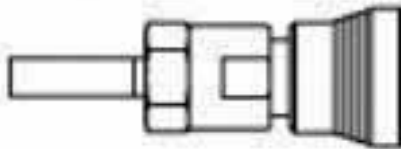
PIL 13K TEU Hudong Shipyard New Builds - LNG tank radar level, temperature and pressure sensor - OBP Spares

Description	Qty / ship
Fuse 0.125A slow, Glass 5x20	3
Tank Gauge Electronics box, 1 channel, ATEX	1
Temp Measuring Block, 4-wire, 3temps, ATEX	1
Temperature sensor (20m)	1

PIL 13K TEU Hudong Shipyard New Builds - LNG tank radar level, temperature and pressure sensor - Standard Spares

Spare Parts

Name	No	Sketch	Quant.	Location
BTT 5110 Box spanner temp tool		 Width of Jaws=12 mm	1	TMS
9150021-048				
VPG 5110 Vapor pressure sensor gasket.			1	TGU
6853501-602				

Purge Nozzle Connection		1	TMS
6853492-731			

46. NITROGEN GENERATOR SYSTEM – AIR PRODUCTS

(OBP SPARE)

1 Year Operational Spare Parts List

Quotation: 15020 System type: 2 off NC1.3-1003
 Date: 04.Aug.25 1x Sperre S37W
 Project: 13k TEU PIL Running hours: 2000 h/year

Generator (1x NC1.3-1816 + 1.3 NC1.3-1003)

Tag No.	Qty.	Replacement cycle	Total Qty.	Description
F-8.11	2	6 months	2	Filter Element. Coalescing, APAS30621-IF
F-8.12	2	6 months	2	Filter Element. Coalescing, APAS30621-PF
F-8.13	2	6 months	2	Filter Element. Coalescing, APAS30621-AF

Feed air compressor (2x Sperre S37W)

Tag No.	Qty.	Replacement after:	Total Qty.	Description
K-8.01	2	500 running houers	2	Sperre Service Kit A

47. FGSS HYDRAULIC VALVE REMOTE CONTROL SYSTEM – HANLA IMS

(OBP SPARE)



PAGE

PROJECT
NAME

Hudong Zhonghua
H1933A/34A/35A/36A/37A

HLS-CHZ2087-OBP-0001

NO.	PART NO.	DESCRIPTION		MAKER	SPARE Q'TY	UNIT PRICE	TOTAL	REMARK
		SPECIFICATION	SKETCH					
1 - 1	VR02577	Return filter element (10 micron)		INTERNORMEN	1	\$ 430	\$ 430	For Hydraulic Power Unit
1 - 2	VR02574	Pressure filter element (10 micron)		STAUFF	1	\$ 280	\$ 280	For Solenoid Valve Cabinet
1 - 3	VR02683	Solenoid valve - 4/2 Way (N.C type, AC220V 60Hz)		HANLA IMS	1	\$ 380	\$ 380	"
1 - 4	VR02684	Solenoid valve - 4/3 Way (S. center type, AC220V 60Hz)		HANLA IMS	2	\$ 380	\$ 760	"
2- 1	VR02787	Emergency hand pump Connector		HANLA IMS	2	\$ 110	\$ 220	For Actuator
2 - 2	VR18245	Isolating valve cartridge		HANLA IMS	3	\$ 130	\$ 390	"
3 - 1	VR00508	Seal kit for HLT-35	-	HANLA IMS	1	\$ 140	\$ 140	"
3 - 2	VR16953	Seal kit for HLT-35 (Low temp.)	-	HANLA IMS	1	\$ 290	\$ 290	
3 - 3	VR00509	Seal kit for HLT-70	-	HANLA IMS	1	\$ 150	\$ 150	"
3 - 4	VR16954	Seal kit for HLT-70 (Low temp.)	-	HANLA IMS	1	\$ 300	\$ 300	"
3 - 5	VR00510	Seal kit for HLT-135	-	HANLA IMS	1	\$ 180	\$ 180	"
3 - 6	VR00511	Seal kit for HLT-300	-	HANLA IMS	1	\$ 190	\$ 190	
3 - 7	VR16956	Seal kit for HLT-300 (Low temp.)	-	HANLA IMS	1	\$ 330	\$ 330	
						AMOUNT	\$ 4,040	

48. LNG TANK INSULATION – TI

(OBP SPARE)



Owner Benefit

To: PACIFIC INTERNATIONAL LINES (PTE) LTD.

Subject: Owner Benefit Proposal for PIL 13000TEU Container vessel Project H1933A-37A

Ti China co., Ltd will provide the following benefit during the Vessel's Operational if Ti China Co Ltd is selected for the LNG Tank Insulation Maker. We confirmed the Owner benefits package to the shipowner, covering the following terms and conditions:

1. Ti can provide 3 years warranty and 2 years as an option in case there is quality issue reported within the first 3 years.
2. Ti will be responsible for the local transportation and accommodation expenses of the owner's on-site personnels in Dalian & Tianjin before the delivery of B tank.
3. Ti will offer the repair kit onboard and provide training for the urgent issue (Material below included in the repair kit).

 REPAIR KIT				
Information: For Panel System				
No	Chinese Name	English Name	Quantity	Unit
1	丁基胶带	Butyl Tape	10	Roll
2	玻璃棉	Glass Wool	5	m ²
3	气凝胶	Nano Thermal Blanket	5	m ²
4	Armaflex	Armaflex	5	m ²
5	5050低温胶水	TIC 5050 Glue	10	kg
6	胶水桶	Platstic bucket	5	Pcs
7	刮板	Stickle	10	Pcs
8	搅拌棒	Stirring rod	1	Pcs



49. PAINT – JOTUN

(OBP SPARE) – SEA STOCK PAINT

H1933A sea stock				
Number	Paint code and name	Unit	Volume	Remarks
1	HARDTOP XP STD2217 RED	L	40	We will supply as per description
2	HARDTOP XP STD1621 BUFF	L	400	include smart package 100L
3	HARDTOP XP WHITE	L	180	include smart package 40L
4	HARDTOP XP STD2343 RB	L	300	include smart package 60L
5	PILOT QD PRIMER PLUS GREY	L	180	We will supply as per description
6	PILOT II WHITE	L	100	We will supply as per description
7	PILOT II STD2343 RB	L	100	We will supply as per description
8	JOTAPRIME S10 ALU RT	L	60	We will supply as per description
9	JOTAPRIME S10 GREY	L	300	We will supply as per description
10	JOTACOTE F60 WHITE	L	20	We will supply as per description
11	SEAQUANTUM X200-3 DR	L	0	This is antifouling, not under seastock. Please re-select
12	SOLVALITT 600	L	20	We will supply Solvalitt Aluminium
13	HARDTOP XP YELLOW	L	40	Please confirm whether STD 258 Yellow?
14	HARDTOP XP GREEN	L	20	Please confirm whether is STD 257/137 Green?
15	HARDTOP XP BLUE	L	20	Please confirm whether for STD 4161 Blue?
16	HARDTOP XP BLACK	L	40	We will supply as per description
17	Black Matt Paint	L	20	
18	Total Litres		1840	
19	Epoxy paint thinner No.17	L	40	
20	Polyurethane paint thinner No.10	L	60	
21	Alkyd paint thinner No.2	L	40	
	Total:140 L			



0258

SIGNAL YELLOW



0257

GREEN



0137

GREEN



4161

SWEET PEACOCK

1	HARDTOP XP STD2217 RED	PU red of TS
2	HARDTOP XP STD1621 BUFF	PU buff of LB
3	HARDTOP XP WHITE	PU white of AC
4	HARDTOP XP STD2343 RB	PU red of DK
5	JOTAPRIME 510 ALU RT	EP primer RT Alu
6	JOTAPRIME 510 GREY	EP topcoat Grey
7	PILOT QD PRIMER PLUS GREY	AK primer of ER
8	PILOT II WHITE	AK topcoat of ER
9	PILOT II STD2343 RB	AK red of ER floor
10	JOTACOTE F60 WHITE	EP funnel inside
11	SOLVALITT 600	HT silver
12	SEAQUANTUM X200-3 DR	Anti-fouling
13	PENGUARD FC	EP topcoat swim po



50. CARGO FIXED FITTING & LASHING – SEC

(OBP SPARE) – TO BE DELIVERED TOGETHER
WITH LOOSE LASHING EQUIPMENT

PIL 13K TEU (H1933A-H1937A) Hudong Shipyard New Builds - Fixed Lashing Equipment - OBP Spare List

S/N	Item Code	Qty / Shipset
1	SF -C1120 .004.56	2
2	SF -C2136A.008.56	2
3	SF -C1124 .002.56	1
4	SF -C1246 .003.56	1
5	SF -C1120 .001.56	6
6	SF -C2136A.010.56	10
7	SF -C1091A.000.56	4
8	WC -C2012 .000.99	20
9	GF -C1245 .000.99	4
10	LAP-C1003G.000.11	20